CFA LEVEL I / DERIVATIVES

# 衍生品

## 65题中英对照精讲

Learn the concepts. Apply them to the supplied questions.

**65**
完整原题 / questions

**9**
学习单元 / units

**22**
代表题 / core examples

### 知识点先行，原题验证
**Concept first, then a worked example**

每个单元先解释概念与公式，再用同主题原题练习。保留全部题干、选项和原参考答案，并补充中英解析、计算步骤及易错点。

Each unit introduces the concept and formulas before the related original questions. All question data, choices and supplied answer letters are retained, with bilingual explanations and worked steps.

易错点 / Take care

快速使用：先看下一页路线，只做 CORE 代表题；然后按错因回到单元补齐练习。题目中的限定与原解析错误另有审校说明。

Start with the next-page route and CORE examples, then complete each unit based on your mistakes. Ambiguous wording and errors in the supplied explanations are explicitly annotated.

题源：1000题(3).pdf，第134-146页；答案：衍生品(1).pdf，第1-65页

按知识点重组的学习讲义 | 编制日期：2026-11-02

# 先学会，再刷完

A fast route through the supplied questions

## 90分钟重点路径 + 全题巩固

### A 90-minute focus route, followed by the full set

先学 9 个单元的知识点，再做标有 CORE 的 22 道代表题。时间紧时按下面顺序推进；想完整掌握时，再完成每单元其余题目。学习时间是安排建议，可按熟悉程度调整。

Read the concept pages, then attempt the 22 questions marked CORE. Use the route below when time is tight, and finish the remaining questions for full practice. The timing is a planning suggestion, adjustable to your familiarity.

| 时间 / Time | 学习任务 / Focus | 原题 / Core questions |
|---|---|---|
| 0-15 min | 合约、市场与对冲风险 / U1-U2 | 13, 53, 14, 19 |
| 15-30 min | 远期与期货 / U3-U4 | 30, 42, 65, 39, 62 |
| 30-45 min | 利率与互换 / U5 | 6, 31, 46 |
| 45-65 min | 期权损益和边界 / U6-U7 | 4, 56, 61, 2, 11 |
| 65-90 min | 平价与二叉树 / U8-U9 | 5, 8, 49, 50, 51 |

## 每题只做四步

### Four steps for every question

1 识别合约和多空方向；2 圈出问的是价格、价值、payoff 还是 profit；3 写出一个公式或一条方向规则；4 再看数字与选项。

1 Identify the contract and position. 2 Mark whether the question asks for price, value, payoff or profit. 3 Write one formula or direction rule. 4 Then use the numbers and choices.

## 三对词，先分清

### Three distinctions to learn first

| 词组 / Terms | 必须分清 / Key distinction |
|---|---|
| Forward price / contract value | 锁定的未来交割价 / 合约今天值多少钱<br>Delivery price / today’s contract value |
| Payoff / profit | 到期支付 / 计入权利金后的净损益<br>Terminal payment / net result including premium |
| Risk-neutral probability / Delta | 定价权重 / 复制所需标的数量<br>Valuation weight / replicating underlying units |

易错点 / Take care

全书题号沿用原题 Q1-Q65。每题同页提供中英题干、选项、答案、解释、计算与易错点；目录及书签可跳转。

Original IDs Q1-Q65 are retained. Each question page contains bilingual wording, choices, answer, explanation, calculations and pitfalls. Use the clickable contents and bookmarks.

# 章节目录与题目地图

Contents and question map

按照知识点学习，而非按原题顺序翻页。各单元先讲概念，再呈现该单元全部原题；原题编号保留，便于回看两份源文件。

Study by concept rather than by the original order. Each unit teaches the ideas before presenting all its supplied questions. Original IDs allow cross-checking with both source files.


| 单元 | 知识点 / Topic | 题数 | 页码 | 原题编号 / Original IDs |
|---|---|---|---|---|
| U1 | 合约、权利与交易市场<br>Contracts, rights and markets | 10 | 4 | Q9, Q12, Q13, Q15, Q17, Q28, Q32, Q35, Q53, Q64 |
| U2 | 对冲用途与风险识别<br>Hedge purposes and risk identification | 6 | 15 | Q10, Q14, Q18, Q19, Q20, Q25 |
| U3 | 无套利、远期价格与持有成本<br>No arbitrage, forward prices and carry | 10 | 22 | Q21, Q24, Q30, Q37, Q38, Q42, Q43, Q55, Q60, Q65 |
| U4 | 期货结算与远期期货差异<br>Futures settlement and forward-futures differences | 5 | 34 | Q22, Q36, Q39, Q40, Q62 |
| U5 | 远期利率、FRA 与互换<br>Forward rates, FRAs and swaps | 6 | 40 | Q6, Q29, Q31, Q33, Q41, Q46 |
| U6 | 期权四种头寸、价值与利润<br>Option positions, payoff and profit | 10 | 48 | Q4, Q16, Q23, Q26, Q48, Q56, Q57, Q58, Q59, Q61 |
| U7 | 期权价值方向与上下界<br>Option value drivers and bounds | 6 | 60 | Q1, Q2, Q3, Q11, Q34, Q63 |
| U8 | 看涨看跌平价与合成头寸<br>Put-call parity and synthetic positions | 6 | 67 | Q5, Q7, Q8, Q27, Q47, Q49 |
| U9 | 二叉树、风险中性概率与复制<br>Binomial trees, risk-neutral probabilities and replication | 6 | 75 | Q44, Q45, Q50, Q51, Q52, Q54 |


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# 合约、权利与交易市场

Contracts, rights and markets

## 先区分：双方的义务，还是买方的权利？

### Start with obligations versus rights

远期、期货和普通互换属于远期承诺：双方必须按合约履行。期权属于或有索取权：买方拥有权利，卖方承担被行权后的义务。

Forwards, futures and plain swaps are forward commitments. Options are contingent claims: the buyer holds a right and the writer has an obligation if exercised.

| 工具 / Instrument | 核心特征 / Key feature | 期初 / At inception |
|---|---|---|
| 远期 Forward | 约定未来买卖；通常场外<br>Agreed future trade; usually OTC | 公允定价时价值为零<br>Zero value if fairly priced |
| 期货 Futures | 标准化、交易所、每日结算<br>Standardized, exchange traded, daily settlement | 通常需保证金<br>Margin is usually required |
| 互换 Swap | 交换一系列现金流<br>Exchange a series of cash flows | 标准平价合约通常为零<br>A standard par swap starts at zero |
| 期权 Option | 买方有选择权；损益不对称<br>Buyer has discretion; asymmetric payoff | 买方向卖方付权利金<br>Buyer pays a premium |

## Call 与 Put：买方到底可以做什么？

### What can an option holder do?

Call 买方有权按执行价买入；Put 买方有权按执行价卖出。Long/short 表示合约买方/卖方，并不直接等于看涨/看跌：买入 put 通常受益于标的下跌。

A call holder may buy at the strike; a put holder may sell at the strike. Long/short describes the position, not simply a bullish/bearish view: a long put generally benefits from a fall.

## 场外、交易所和嵌入式期权

### OTC, exchanges and embedded options

非标准数量、期限和交割日优先考虑 OTC，交易对手通常是交易商。交易所合约通常更标准、更透明、交易成本较低。可赎回债券含发行人的赎回期权；单独的 put 或 futures 本身是衍生品，不因此叫“嵌入式”。

OTC markets suit customized size and dates, with dealers commonly facing end users. Exchange contracts are usually more standardized and transparent, with lower transaction costs. A callable bond embeds the issuer’s call right; a standalone put or future is a derivative in its own right.

易错点 / Take care

做题口令：先判 firm commitment / contingent claim，再判 premium / margin。对应 Q13、Q35、Q53。

Exam trigger: classify the commitment or claim, then distinguish premium from margin. Apply to Q13, Q35 and Q53.

# 原题 Q09

合约、权利与交易市场 | Contracts, rights and markets

下列哪项最准确?

Which of the following is most accurate?

**A** 远期合约在有组织的交易所交易。
A forward contract is traded on an organized exchange.

**B** 远期合约比期货合约更透明。
Forward contracts are more transparent than futures contracts.

**C** 远期合约买方同意在未来某一日期按固定价格买入标的资产。
The buyer of a forward contract agrees to buy the underlying asset at a fixed price on a future date.

## 参考答案 / Supplied answer C

远期合约双方今天约定未来交易的交割价格，买方承担届时按约买入的义务，所以 C 正确。标准远期通常在场外协商，而期货在交易所交易且更标准化、更透明，因此 A、B 将远期与期货混淆。

A forward fixes today the delivery price for a future transaction, and the buyer is obligated to buy on the agreed terms. Standard forwards are generally negotiated over the counter, whereas futures are exchange-traded, standardized, and more transparent. A and B confuse these features.

易错点 / Take care

Forward buyer 有买入义务；call buyer 则拥有可以放弃的买入权利。

A forward buyer has an obligation to buy; a call buyer has a right that may be left unexercised.

# 原题 Q12

合约、权利与交易市场 | Contracts, rights and markets

下列哪种工具的结构中最可能含有嵌入式衍生品?

Which of the following most likely has an embedded derivative in its structure?

**A** 看跌期权。
A put option

**B** 可赎回债券。
A callable bond

**C** 期货合约。
A futures contract

## 参考答案 / Supplied answer B

嵌入式衍生品是写进另一种金融工具条款中的衍生权利。可赎回债券含有发行人的提前赎回权，可理解为普通债券附带了债券持有人卖出的赎回期权。单独的 put 和 futures 本身是独立衍生品。

An embedded derivative is a derivative feature incorporated into another financial instrument. A callable bond contains the issuer's right to redeem early and can be viewed from the investor's side as a straight bond with a written call. A standalone put or futures contract is a separate derivative.

易错点 / Take care

Embedded 指“嵌在另一个工具里”，不是“这项工具本身是衍生品”。

Embedded means incorporated into another instrument, rather than merely being a derivative itself.

# 原题 Q13

代表题 / CORE

合约、权利与交易市场 | Contracts, rights and markets

下列哪种衍生品最可能要求在合约开始时支付款项?

Which of the following derivatives most likely requires a payment to be made at the initiation of the contract? A(n):

**A** 互换。
swap

**B** 期权。
option

**C** 远期。
Forward

## 参考答案 / Supplied answer B

期权买方向卖方支付权利金，以购买可选择行权的权利，所以通常在初始时需要支付款项。按当前市场条件公平定价的普通远期和互换，其初始合约价值通常为零，通常不需要支付合约购买价。

An option buyer pays a premium to acquire the right to choose whether to exercise, so an upfront payment is normally required. Ordinary forwards and swaps entered at fair market terms normally have zero initial contract value and require no purchase price at inception.

易错点 / Take care

初始价值为零不等于绝不需要抵押品；保证金与购买期权的权利金用途不同。

Zero initial contract value does not mean collateral can never be required. Margin and an option premium serve different purposes.

# 原题 Q15

合约、权利与交易市场 | Contracts, rights and markets

下列哪种衍生品合约最适合归类为或有索取权?

Which of the following derivative contracts is best described as a contingent claim?

**A** 互换合约。
A swap contract

**B** 远期合约。
A forward contract

**C** 期权合约。
An option contract

## 参考答案 / Supplied answer C

期权是否产生行权收益取决于到期状态及持有人的行权权利，所以属于或有索取权。远期和普通互换属于确定承诺：双方已经承诺按合约履行，不能仅因为交易变得不利而选择不履约。

An option is a contingent claim because exercise payoffs depend on the terminal state and the holder's exercise right. Forwards and ordinary swaps are firm commitments: each side has committed to perform and cannot simply decline because the position becomes unfavorable.

易错点 / Take care

Contingent claim 的关键是期权性质；不是“合约价值会变化”就属于或有索取权。

The key is the option feature, not simply that the contract value changes.

# 原题 Q17

合约、权利与交易市场 | Contracts, rights and markets

下列哪些衍生品具有非线性的到期收益?

Which of the following derivatives have a non-linear payoff?

**A** 只有或有索取权。
Contingent claims only

**B** 只有远期承诺。
Forward commitments only

**C** 或有索取权及远期承诺两者都有。
Both contingent claims and forward commitments

## 参考答案 / Supplied answer A

标准期权的到期收益包含 max 函数，到执行价处改变斜率，因此属于非线性衍生品。普通远期承诺的到期收益随标的价格作线性变化，例如多头远期为 S_T - K，所以不属于本题所问的非线性类别。

A standard option payoff contains a max function and changes slope at the strike, making it non-linear. Ordinary forward commitments have payoffs that change linearly with the underlying, such as S_T - K for a long forward. Therefore only contingent claims are selected in this classification.

Long forward payoff = $S_T - K$

Long call payoff = $\max(S_T - K, 0)$

易错点 / Take care

非线性是指相对标的价格的收益形状；并非泛指所有估值变量的数学关系。

Non-linear here describes the payoff against the underlying price, rather than every mathematical relationship used in valuation.

# 原题 Q28

合约、权利与交易市场 | Contracts, rights and markets

与场外衍生品相比，交易所交易的衍生品通常具有哪一项特征？

Compared to over-the-counter derivatives, exchange-traded derivatives:

**A** 标准化程度较低。
are less standardized.

**B** 透明度较低。
provide less transparency.

**C** 交易成本较低。
have lower transaction costs.

## 参考答案 / Supplied answer C

交易所合约通常具有标准化条款、公开报价及较集中的流动性，因此通常交易成本较低。A、B 将两类市场的典型特征倒置。场外合约的优势是可以定制数量、标的和结算日期。

Exchange-traded contracts usually have standardized terms, visible prices, and concentrated liquidity, supporting lower transaction costs. A and B reverse the typical comparison. OTC contracts instead offer flexibility to customize the underlying, size, and settlement dates.

易错点 / Take care

这里是市场结构的通常比较，不代表每一笔交易所合约的实际总成本必然更低。

This is a typical market-structure comparison, not a guarantee that every exchange-traded transaction has lower total costs.

# 原题 Q32

合约、权利与交易市场 | Contracts, rights and markets

在场外衍生品市场中，大多数交易通常发生在最终用户与哪一类参与者之间?

In the over-the-counter derivatives market, most transactions occur between end users and:

**A** 交易商。
dealers.

**B** 其他最终用户。
other end users.

**C** 中央对手方。
a central counterparty.

## 参考答案 / Supplied answer A

最终用户通常与专业交易商交易，由交易商报价、提供定制合约并管理其风险。中央对手方主要负责合约清算及风险管理，不是这里所问的一般报价交易对手。部分 OTC 交易可进入中央清算，但这不改变题目对交易执行关系的典型描述。

End users typically transact with dealers that quote prices, customize terms, and manage the resulting exposures. A central counterparty primarily provides clearing and risk management rather than acting as the usual quoting dealer. Some OTC trades are centrally cleared, but that does not change the typical execution relationship asked here.

易错点 / Take care

交易执行对象与清算后的法律交易对手不是完全相同的问题。

The execution counterparty and the legal counterparty after clearing are not necessarily the same question.

# 原题 Q35

合约、权利与交易市场 | Contracts, rights and markets

下列组合包含一家公司的股票，以及以该股票为标的的衍生品。哪一个组合中的衍生品最可能属于用于对冲股票的确定承诺?

The following portfolios contain a company's stock and a derivative on the stock. The portfolio containing a derivative acting as a firm commitment to hedge the stock is most likely:

| 组合 Portfolio | 持仓 Securities |
|---|---|
| 1 | 股票 + 期货空头 / Stock and a short futures position |
| 2 | 股票 + 看涨期权空头 / Stock and a short call option position |
| 3 | 股票 + 认股权证空头 / Stock and a short warrant position |

**A** 组合 1。
Portfolio 1.

**B** 组合 2。
Portfolio 2.

**C** 组合 3。
Portfolio 3.

## 参考答案 / Supplied answer A

期货要求双方按约履行，是确定承诺，所以股票加期货空头符合题意。看涨期权和认股权证属于或有请求权，是否执行取决于条件和持有人选择。卖出看涨期权虽然可以收取权利金，却并不等同于用期货空头锁定股票的出售价格。

A futures contract obligates both parties and is therefore a firm commitment; stock plus short futures fits the question. Calls and warrants are contingent claims whose exercise depends on conditions and the holder's choice. A short call generates premium income but does not lock in the stock's sale price in the same way as short futures.

易错点 / Take care

covered call 仍承担股票大幅下跌风险，不能把收到权利金误认为完整的价格对冲。

A covered call still bears substantial downside in the stock; premium income is not a complete price hedge.

# 原题 Q53

代表题 / CORE

合约、权利与交易市场 | Contracts, rights and markets

某终端使用者希望对特定标的风险敞口进行对冲，但所需合约的规模及结算日期均非标准化。其最可能在哪一种市场交易？

An end user seeking to hedge a specific underlying exposure having non-standard size and settlement dates would most likely trade on a(n):

**A** 期货市场。
futures market.

**B** 场外衍生品市场。
over-the-counter derivative market.

**C** 交易所衍生品市场。
exchange-traded derivative market.

## 参考答案 / Supplied answer B

场外市场允许交易双方协商名义本金、到期日及结算条款，更适合特定而非标准化的风险敞口。交易所合约通常使用标准化规模和到期日，未必能精确匹配本题需求。

The OTC market allows parties to negotiate notional amounts, maturities, and settlement terms, making it suitable for a specific non-standard exposure. Exchange-traded contracts generally have standardized sizes and expiry dates and may not match this need precisely.

易错点 / Take care

定制需求对应OTC；不要因为“对冲”二字就自动选期货。

Customization points to OTC. The mere presence of a hedging objective does not automatically imply futures.

# 原题 Q64

合约、权利与交易市场 | Contracts, rights and markets

期权合约中的哪一方拥有按行权价卖出标的股票的权利?

Which party in an option contract has the right to sell the underlying stock at the exercise price?

**A** 看涨期权买方。
The buyer of a call option

**B** 看跌期权买方。
The buyer of a put option

**C** 看跌期权卖方。
The seller of a put option

## 参考答案 / Supplied answer B

看跌期权赋予买方按行权价卖出标的的权利，是否行权由买方决定。相应地，看跌期权卖方在被行权时负有按行权价买入标的的义务。看涨期权买方则拥有买入权。

A put gives its buyer the right to sell the underlying at the strike, with exercise chosen by the buyer. If exercised, the put seller is obliged to buy the underlying at the strike. A call buyer instead has the right to buy.

易错点 / Take care

先区分call=买入权、put=卖出权，再区分买方拥有权利、卖方承担义务。

First distinguish call = right to buy and put = right to sell; then distinguish the buyer's right from the seller's obligation.

# 对冲用途与风险识别

Hedge purposes and risk identification

## 先看“什么在错配”

### Identify what fails to match

| 风险 / Risk | 关键词与含义 / Trigger and meaning | 原题 / Q |
|---|---|---|
| 基差风险<br>Basis risk | 价格或价值变动不完全匹配<br>Derivative and hedged exposure do not move as expected | 14, 18 |
| 流动性风险<br>Liquidity risk | 现金流时间错配、无钱补保证金<br>Cash-flow timing mismatch or inability to fund margin | 25 |
| 对手信用风险<br>Counterparty credit risk | 另一方不履行付款义务<br>The other party fails to perform | 22 |
| 系统性风险<br>Systemic risk | 高杠杆、传染与市场压力<br>Leverage, contagion and market stress | 20 |

经济上“最终能对冲”，不等于过程中“现金够用”。例如期货先发生保证金流出、被对冲业务日后才收款，仍有资金周转风险。

An eventual economic offset does not guarantee sufficient cash along the way. Margin outflows can arrive before receipts from the hedged business.

## 三种对冲：看被保护的项目

### Three hedge labels: identify the exposure

| 对冲类型 / Designation | 本套题的识别方式 / Recognition rule |
|---|---|
| 公允价值对冲<br>Fair value hedge | 已有存货的价格下跌风险；Q10<br>Changes in the value of existing inventory; Q10 |
| 现金流对冲<br>Cash flow hedge | 未来预计交易的现金流不确定；Q19 的干扰项<br>Variability in forecast transaction cash flows |
| 境外经营净投资对冲<br>Net investment hedge | 境外经营权益的外汇风险；Q19C<br>FX risk in the equity of a foreign operation; Q19C |

对冲降低某项暴露，仍可能留下基差、信用和流动性风险。衍生品通常有助于价格发现并降低交易成本；过度杠杆和风险承担也可能加剧市场压力。

Hedging can reduce one exposure while leaving basis, credit or liquidity risk. Derivatives can support price discovery and reduce transaction costs, while excessive leverage can amplify stress.

易错点 / Take care

辨别 Q14 与 Q25：value divergence 对应 basis；cash-flow timing divergence 在本题口径下对应 liquidity。

For Q14 versus Q25: value divergence points to basis risk; cash-flow timing divergence points to liquidity risk in this question set.

# 原题 Q10

对冲用途与风险识别 | Hedge purposes and risk identification

某大宗商品生产商预期未来价格下降，因此通过远期合约卖出现有存货。这属于哪一种套期保值?

A commodities producer selling its inventory forward in anticipation of lower prices in the future is an example of a:

**A** 公允价值套期。
fair value hedge.

**B** 现金流量套期。
cash flow hedge.

**C** 净投资套期。
net investment hedge.

## 参考答案 / Supplied answer A

题目保护的是已经持有的存货，其公允价值可能随商品价格下跌而减少。卖出远期的价值在价格下跌时提高，可以抵销这项价值变化，因此按题目指定的用途归为公允价值套期。若保护的是尚未发生的预测销售现金流，分类才通常是现金流量套期。

The exposure being protected is existing inventory, whose fair value can fall with commodity prices. A short forward gains value as those prices fall, offsetting the inventory value change, so the stated use is a fair value hedge. Hedging the cash flows of a forecast sale would generally be a cash flow hedge instead.

易错点 / Take care

先找被套期项目：existing inventory 对应公允价值；forecast transaction 对应现金流量。

Identify the hedged exposure first: existing inventory points to fair value; a forecast transaction points to cash flows.

# 原题 Q14

代表题 / CORE

对冲用途与风险识别 | Hedge purposes and risk identification

基差风险最适合描述为下列哪种情况?

Basis risk is best described as a(n):

**A** 投资者由于缺乏资金而无法满足追加保证金要求。
investor's inability to meet a margin call due to a lack of funds.

**B** 衍生品与其标的之间预期价值可能发生偏离。
potential divergence between the expected value of a derivative and its underlying.

**C** 衍生品与标的交易的现金流发生时点不一致。
divergence in the cash flow timing of a derivative versus that of an underlying transaction.

## 参考答案 / Supplied answer B

基差风险关注的是衍生品与被套期头寸的价格或价值变化不能完全匹配，因此套期损益无法完全抵销。A 的重点是缺乏即时现金，C 的重点是收付款时点不一致，二者都指向流动性风险。

Basis risk concerns an imperfect match between the price or value changes of the derivative and the hedged exposure, so hedge gains may not fully offset underlying losses. A focuses on insufficient immediate cash, while C focuses on mismatched cash-flow timing. Both relate to liquidity risk.

易错点 / Take care

价值变化对不上是 basis risk；现金到达时间对不上是 liquidity risk。

A mismatch in value changes is basis risk; a mismatch in cash-flow timing is liquidity risk.

# 原题 Q18

对冲用途与风险识别 | Hedge purposes and risk identification

衍生工具与标的或被套期交易的预期价值之间可能发生偏离，最适合称为哪种风险?

The potential divergence between the expected value of a derivative instrument versus an underlying or hedged transaction best describes:

**A** 基差风险。
basis risk.

**B** 流动性风险。
liquidity risk.

**C** 系统性风险。
systemic risk.

## 参考答案 / Supplied answer A

当衍生品与被套期项目的价值变化并不完全一致时，就存在基差风险。这里关键词是价值偏离，而不是现金流时点不匹配，也不是风险扩散至整个金融系统。

Basis risk arises when changes in derivative value do not perfectly match changes in the hedged exposure. The key phrase is divergence in value, not mismatched cash-flow timing or risk spreading through the financial system.

易错点 / Take care

本题与 Q14 考同一知识点：把 expected value divergence 直接对应到 basis risk。

This repeats Q14: link expected value divergence to basis risk.

# 原题 Q19

代表题 / CORE

对冲用途与风险识别 | Hedge purposes and risk identification

关于套期会计的指定类别，下列哪项正确?

With respect to hedge accounting designation types, a:

**A** 用于对冲预测销售的外汇远期属于公允价值套期。
foreign exchange forward to hedge forecasted sales is an example of a fair value hedge.

**B** 用于对冲存货的商品期货属于现金流量套期。
commodity futures contract used to hedge inventory is an example of a cash flow hedge.

**C** 用于抵销境外经营权益所承担的汇率风险的货币远期属于净投资套期。
currency forward to offset the foreign exchange risk of equity of a foreign operation is an example of a net investment hedge.

## 参考答案 / Supplied answer C

境外经营的权益属于对海外净投资的敞口，因此对冲其汇率风险属于净投资套期。A 的预测销售对应未来现金流，通常属于现金流量套期；B 按本题对现有存货价值的保护方式，应归为公允价值套期。

The equity of a foreign operation represents exposure to a net investment abroad, so hedging its exchange-rate risk is a net investment hedge. Forecast sales in A concern future cash flows and normally imply a cash flow hedge. Under the stated use in B, protecting the value of existing inventory is a fair value hedge.

易错点 / Take care

不要按使用了 FX forward 还是 futures 来分类，要按究竟保护哪项风险敞口来分类。

Classify the hedge by the exposure being protected, not by whether the instrument is an FX forward or a futures contract.

# 原题 Q20

对冲用途与风险识别 | Hedge purposes and risk identification

下列关于衍生品的说法，哪项最准确?

Which of the following statements about derivatives is most accurate?

**A** 衍生品会降低标的资产的价格发现效率。
Derivatives reduce the efficiency of price discovery for the underlying

**B** 衍生品的交易成本高于标的资产的交易成本。
Transaction cost of derivatives are greater than the transaction cost of the underlying

**C** 衍生品市场中的过度风险承担及杠杆使用可能加剧市场压力。
Excessive risk taking and use of leverage in derivative markets may contribute to market stress

## 参考答案 / Supplied answer C

衍生品可以帮助价格发现，也通常使调整风险敞口所需的交易成本和初始现金投入较低。与此同时，过度杠杆会放大损失、追加保证金压力和被迫平仓的影响，因此 C 正确。成本较低或有套期功能，并不代表整体风险必然较低。

Derivatives can support price discovery and often permit exposure adjustments with lower transaction costs and initial cash outlays. Excessive leverage can nevertheless magnify losses, margin pressure, and forced liquidation, making C correct. Lower cost or the ability to hedge does not guarantee lower overall risk.

易错点 / Take care

Benefits 与 risks 可以同时存在；不要把能对冲风险误读为不会放大风险。

Benefits and risks can coexist. The ability to hedge does not prevent derivatives from amplifying risk.

# 原题 Q25

对冲用途与风险识别 | Hedge purposes and risk identification

衍生工具与其标的资产的现金流发生时间可能不一致，最适合描述哪一种风险?

The potential divergence between the cash flow timing of a derivative instrument versus its underlying best describes:

**A** 基差风险。
basis risk

**B** 流动性风险。
liquidity risk.

**C** 系统性风险。
systemic risk.

## 参考答案 / Supplied answer B

本题强调现金流的时间错配，因此是流动性风险。例如，期货亏损可能今天就要补保证金，但被对冲资产的收益要到未来出售时才收到。最终对冲可能有效，期间仍可能因缺少现金而无法维持头寸。

The key phrase is a mismatch in cash-flow timing, which creates liquidity risk. For example, futures losses may require cash margin today while gains on the hedged asset are only realized when it is sold later. A hedge can work economically at maturity yet create an interim funding shortage.

易错点 / Take care

金额或价格联动不匹配看 basis risk；到账或付款时间不匹配看 liquidity risk。

A mismatch in price movements points to basis risk; a mismatch in payment timing points to liquidity risk.

# 无套利、远期价格与持有成本

No arbitrage, forward prices and carry

## 同样的未来现金流，今天应有同样的价格

### Identical future cash flows imply the same price

借钱买入现货并持有至到期，可以复制某些远期头寸。若两条路径成本不同，买便宜的一边、卖贵的一边会形成套利。因此定价不需要先预测真实世界的未来股价。

Borrowing to buy and carry the underlying can replicate a forward exposure. Different costs for equivalent cash flows would allow arbitrage. Pricing therefore does not start with a forecast of the actual future stock price.

| 符号 / Symbol | 含义 / Meaning |
|---|---|
| S0, St, ST | 当前、时点 t、到期的标的现货价<br>Spot prices now, at t, and at maturity |
| F0(T), K | F0(T) 是期初公允交割价格；订约后旧合约的 K 固定<br>Initial fair delivery price; K stays fixed for the existing contract |
| Vt, N, τ | 合约价值、本节标的单位数、剩余年数 T-t<br>Contract value, underlying units in this section, and remaining years |

## 无收益、无额外持有成本的基本式

### Baseline: no income or additional carrying cost

$F_0(T) = S_0 \times (1 + r)^T$

$V_0 = 0$ when $K = F_0(T)$

$V_t,long = N \times [S_t - K / (1 + r)^(T-t)]$

$V_T,long = N \times (S_T - K)$; $V_T,short = N \times (K - S_T)$

“合约价值为零”不代表交割价格为零。Q30 问 V0；Q55 问到期卖方价值；Q65 固定旧 K 和现货价，利率上升使 K 的现值下降，远期多头价值上升。

Zero contract value does not mean a zero delivery price. Q30 asks about V0, Q55 about the seller’s terminal value, and Q65 holds the old K and spot price fixed: a higher rate lowers PV(K) and raises the long forward’s value.

易错点 / Take care

Q24 的无风险复制组合必须赚取无风险收益率；Q43 考一价定律。这里默认可复制、无交易摩擦及一致的借贷利率。

Q24 requires a riskless replicating portfolio to earn the risk-free rate; Q43 tests the law of one price. The model assumes replicability, no frictions and consistent financing rates.

# 无套利、远期价格与持有成本

No arbitrage, forward prices and carry

## 把融资、额外成本和持有收益分开

### Separate financing, extra costs and holding benefits

$F_0(T) = [S_0 + PV(costs) - PV(benefits)] \times (1 + r)^T$

If a cost CT is paid at T: $F_0(T) = S_0 \times (1 + r)^T + CT$

Continuous rates: $F_0(T) = S_0 \times \exp[(r + u - q - y) \times T]$

r 是融资利率；u 是仓储等成本率；q 是股息或现金收益率；y 是便利收益率。连续复利公式中的各项都要使用一致的连续复利口径。

Here r is financing, u a carrying-cost rate, q a dividend or income yield, and y a convenience yield. Rates in the exponential formula must use consistent continuous-compounding conventions.

| 因素增加 / Increase in | 远期价格 / Forward price | 直觉 / Intuition |
|---|---|---|
| 融资利率 r / Financing rate | 上升 / Higher | 持有现货融资更贵<br>More expensive to finance ownership |
| 仓储成本 u / Storage cost | 上升 / Higher | 持有现货成本增加<br>Ownership becomes more costly |
| 股息 q / Dividend yield | 下降 / Lower | 现货持有人可取得收入<br>Owner receives income |
| 便利收益 y / Convenience yield | 下降 / Lower | 现货可用于生产或避免短缺<br>Physical inventory offers operational benefits |

## Q42：付款时点决定是否需要复利

### Q42: timing determines whether a cost is compounded

T = 182 / 365

$F_0 = 120 \times 1.03^{(182/365)} + 5 = 126.78$

仓储费 5 美元在到期支付，所以直接加在终值上；不要把这 5 美元再复利一次。题中使用年复利及 365 天年基准。

The USD 5 storage cost is paid at maturity, so add it directly to the terminal value. Do not compound it again. This question specifies annual compounding and a 365-day year.

易错点 / Take care

Q38 的 net cost of carry 指融资以外的成本与收益净额为零，所以仍保留融资：F0=S0(1+r)^T。若把“总持有成本”定义为连 r 也包含且总和为零，才会得到 F0=S0。

In Q38, zero net carry means extra benefits and costs cancel; financing remains, so F0=S0(1+r)^T. If total carry including r were zero, the result would instead be F0=S0.

# 原题 Q21

无套利、远期价格与持有成本 | No arbitrage, forward prices and carry

下列哪类资产最可能具有便利收益?

Which of the following asset classes is most likely to have a convenience yield?

**A** 大宗商品。
Commodities

**B** 利率。
Interest rates

**C** 外汇。
Foreign exchange

## 参考答案 / Supplied answer A

便利收益是实际持有实物商品所带来的非现金好处，例如随时可用的库存能支持生产或满足即时交货需求。仅持有商品的远期或期货无法完全获得这种即时使用的利益，所以这一概念最典型地用于商品。

Convenience yield is the non-cash benefit from physically holding a commodity, such as having inventory available for production or immediate delivery. A commodity forward or futures position does not fully provide that immediate-use benefit. The concept is therefore most closely associated with commodities.

易错点 / Take care

Convenience yield 是持有实物的收益；storage cost 是持有实物的成本，方向相反。

Convenience yield is a benefit of physical ownership; storage cost is a carrying cost, so their pricing effects differ.

# 原题 Q24

无套利、远期价格与持有成本 | No arbitrage, forward prices and carry

衍生品通常通过构建包含标的资产与衍生品的对冲组合来定价；这个组合必须提供哪一种收益率？

Derivatives are typically priced by forming a hedge involving the underlying asset and a derivative such that the combination must pay the:

**A** 无风险利率。
risk-free rate.

**B** 股息收益率。
dividend yield.

**C** 便利收益率。
convenience yield.

## 参考答案 / Supplied answer A

如果一个适当构建的组合已消除风险，它必须赚取无风险利率，否则可以通过借贷和反向交易获利。令组合的收益等于无风险收益，就能反推出衍生品的无套利价格。股息率和便利收益率可以影响持有成本，但不是无风险组合的必要收益率。

A properly constructed riskless portfolio must earn the risk-free rate; otherwise borrowing, lending, and offsetting trades create an arbitrage. Equating the hedged return to the risk-free return determines the derivative price. Dividend and convenience yields can affect carrying benefits but do not replace the required riskless return.

易错点 / Take care

无套利定价不需要先预测投资者真实的风险偏好或标的预期收益率。

No-arbitrage pricing does not first require an estimate of investors' actual risk preferences or the underlying's expected return.

# 原题 Q30

代表题 / CORE

无套利、远期价格与持有成本 | No arbitrage, forward prices and carry

远期合约在订立时的价值最有可能等于多少?

The value of a forward contract at initiation is most likely equal to:

**A** 零。
zero.

**B** 现货价格减远期价格。
the spot price minus the forward price.

**C** 远期价格减现货价格。
the forward price minus the spot price.

## 参考答案 / Supplied answer A

标准远期会把交割价设在使多空双方初始价值均为零的水平，因此通常无需支付权利金。远期价格 F0 是未来约定交割价，远期价值 V0 是今天这份合约的价值，两者不是同一概念。现货与交割价必须先调整时间价值及持有收益，不能直接相减。

A standard forward sets its delivery price so that both sides have zero value at inception and no option-like premium is paid. F0 is the agreed future delivery price, while V0 is the contract's value today. Spot and delivery prices cannot simply be subtracted without adjusting for time value and carrying benefits.

> For a standard at-market forward: $V_0 = 0$.

易错点 / Take care

交割价可以是 285，合约初始价值仍可以是 0；价格与价值必须分开。

A delivery price of 285 is fully consistent with an initial contract value of zero: price and value are different.

# 原题 Q37

无套利、远期价格与持有成本 | No arbitrage, forward prices and carry

某股票的股息收益率为 3%，现货价格为 50 美元。若年化无风险利率为 5%，该股票 6 个月的远期价格最接近多少?

A stock with a dividend yield of 3% is trading in the spot market at $50. If the annual risk-free rate is 5%, the 6-month forward price of the stock is closest to:

**A** 49.50 美元。
$49.50.

**B** 50.50 美元。
$50.50.

**C** 51.27 美元。
$51.27.

## 参考答案 / Supplied answer B

远期买方直到到期才付款，需考虑融资成本 r；但他没有持有现货期间的股息，因此需减去股息收益率 q。净持有成本率约为 5%-3%=2%，半年对应约 1% 的增幅。计算得到约 50.50 美元，选择 B。

The forward buyer delays payment until delivery, creating a financing cost r, but misses the dividends received by a spot holder, so q must be deducted. The net carrying rate is about 5%-3%=2%, or about 1% over half a year. The resulting forward price is approximately $50.50, giving B.

> T = 6/12 = 0.5; r-q = 0.05-0.03 = 0.02.
>
> $F_0$ = 50 × exp((0.05-0.03)*0.5) = 50.5025, approximately 50.50.

### 易错点 / Take care

不能把股息率加到无风险利率上；股息是持有现货的好处，会降低远期价格。

Do not add the dividend yield to the risk-free rate: dividends benefit the spot holder and reduce the forward price.

### 审校说明 / Wording and scope

原答案使用连续复利公式；题干未明确复利约定，但常见离散近似也对应 50.50 这一选项。

The source solution uses continuous compounding. The stem does not specify a convention, but common discrete approximations also select $50.50.

# 原题 Q38

无套利、远期价格与持有成本 | No arbitrage, forward prices and carry

如果净持有成本为零，商品远期价格最可能与哪一个金额相等、较低或较高？比较基准为：商品现货价格按无风险利率复利增长至合约到期的金额。

If the net cost of carry is zero, the forward price of a commodity is most likely:

**A** 低于上述比较基准。
less than the commodity's spot price compounded at the risk-free rate over the life of the contract.

**B** 等于上述比较基准。
equal to the commodity's spot price compounded at the risk-free rate over the life of the contract.

**C** 高于上述比较基准。
greater than the commodity's spot price compounded at the risk-free rate over the life of the contract.

## 参考答案 / Supplied answer B

本题答案把净持有成本用于描述融资以外的持有成本与持有收益之净额；该净额为零时，它们互相抵销。融资的时间价值仍然存在，因此远期价格等于现货价格按无风险利率累积至到期的金额。选择 B，不能直接推成 F0=S0。

In this solution, net cost of carry refers to non-financing carrying costs and benefits. When those items offset, the financing time value still remains. Therefore the forward price equals the spot price compounded at the risk-free rate, giving B rather than simply F0=S0.

> $F_0 = [S_0 + PV(costs) - PV(benefits)] \times (1+r)^T$.
>
> If $PV(costs) = PV(benefits)$: $F_0 = S_0 \times (1+r)^T$.

易错点 / Take care

先看 net cost of carry 是否包含融资成本；不同资料的用词范围可能不同。

First check whether net cost of carry includes financing: terminology can differ across sources.

审校说明 / Wording and scope

原解以 benefits-minus-costs 表示净项，再从 S0 中减去该项；本讲义改用 costs-minus-benefits 的加法写法，经济含义相同。若别的资料把 r 也计入总净持有成本，则总净成本为零才会得到 F0=S0。

The source subtracts a benefits-minus-costs term from S0; this guide equivalently adds costs-minus-benefits. If another source includes r in total net carry, zero total net carry would instead imply F0=S0.

# 原题 Q42

代表题 / CORE

无套利、远期价格与持有成本 | No arbitrage, forward prices and carry

分析师收集到：原油现货价格每桶 120 美元；无风险年利率为 3%，按年复利；期货距离结算有 182 天；每桶储存成本为 5 美元，并在期货到期时支付。按每年 365 天计，原油期货价格每桶最接近多少？

An analyst gathers the following information: the current spot price of crude oil is $120 per barrel; the risk-free rate is 3% with annual compounding; a futures contract has 182 days until settlement; and the storage cost is $5 per barrel, payable at the end of the futures contract. Based on 365 days per year, the futures price per barrel of crude oil is closest to:

**A** 126.78 美元。
$126.78.

**B** 126.86 美元。
$126.86.

**C** 126.93 美元。
$126.93.

## 参考答案 / Supplied answer A

先把今天购买原油的 120 美元按年复利累积 182/365 年，再加上到期才支付的 5 美元储存费。储存费已是到期现金流，不能再对这 5 美元计息。总价约为 126.78 美元，选择 A。

First compound the $120 spot purchase cost for 182/365 of a year, then add the $5 storage charge paid at maturity. Because the storage fee is already a maturity cash flow, it must not be compounded again. The total is about $126.78, giving A.

> T = 182/365.
>
> $F_0$ = 120 × (1.03)^(182/365) + 5 = 126.7818, approximately 126.78.

易错点 / Take care

储存费期初支付要计融资成本；期末支付直接加在到期金额上。

Storage paid upfront must be financed; storage paid at maturity is added directly to the terminal amount.

审校说明 / Wording and scope

题目采用标准无套利持有成本模型，并未给便利收益等其他持有收益。

The question uses the standard carrying-cost model and supplies no convenience yield or other carrying benefit.

# 原题 Q43

无套利、远期价格与持有成本 | No arbitrage, forward prices and carry

大多数衍生品定价模型建立在哪一个基础之上?

Most derivatives pricing models are established on the foundation that:

**A** 存在套利机会。
arbitrage opportunities exist.

**B** 一种衍生品只能有一个价格。
only one price for a derivative exists.

**C** 通过推断标的资产价格来确定衍生品价格。
the underlying asset price is inferred to determine the derivative price.

## 参考答案 / Supplied answer B

核心是无套利与一价定律：如果相同未来现金流可以由两种组合复制，两者今天应具有相同价格。定价模型以已知的标的价格和可复制现金流推导衍生品价格，并不以存在套利为前提。题目中的“一个价格”指相同条款、时点与条件下的理论价格。

The foundation is no arbitrage and the law of one price: portfolios with identical future cash flows should have the same current price. Pricing starts from the observed underlying and replicating cash flows, rather than assuming arbitrage exists. One price means a theoretical value for the same terms, time, and conditions.

易错点 / Take care

现实报价可能有买卖价差；一价定律是定价框架，并不是说屏幕上永远只有一个报价。

Real markets can have bid-ask spreads; the law of one price is a pricing principle, not a claim that only one displayed quote exists.

# 原题 Q55

无套利、远期价格与持有成本 | No arbitrage, forward prices and carry

一份远期合约的条款如下。到期时现货价格为282美元，该合约卖方的价值是多少?

A forward agreement has the following terms. At expiration, if the spot price is $282, the value to the seller is:

| 项目 / Item | 数值 / Value |
|---|---|
| 签约时现货价格 / Spot price at inception | $275 |
| 远期交割价格 / Forward price | $285 |
| 股票数量 / Number of shares | 2,000 |

**A** -$6,000.

**B** $6,000.

**C** $14,000.

## 参考答案 / Supplied answer B

远期卖方在到期时按约定价格285美元卖出，而当时市场价格为282美元，所以每股获利3美元。乘以2,000股，卖方到期价值为6,000美元。签约时的现货价格275美元不用于计算这份远期合约的到期支付。

The forward seller delivers at the agreed $285 price when the market price is $282, gaining $3 per share. For 2,000 shares, the seller's expiration value is $6,000. The initial $275 spot price does not enter the forward contract's expiration payoff.

> Short forward payoff = $N \times (K - S_T)$
>
> Payoff = $2{,}000 \times (285 - 282) = 6{,}000$

易错点 / Take care

卖方用K - S_T，买方才用S_T - K；不要用初始现货价275计算远期盈亏。

A seller uses K - S_T; a buyer uses S_T - K. Do not calculate the forward payoff using the initial spot price of $275.

# 原题 Q60

无套利、远期价格与持有成本 | No arbitrage, forward prices and carry

其他条件不变，支付股息的股票，其持有成本如何?

All else being equal, the cost of carry on a dividend-paying stock is:

**A** 低于不支付股息股票的持有成本。
lower than the cost of carry on a stock with no dividends.

**B** 与不支付股息股票的持有成本相同。
the same as the cost of carry on a stock with no dividends.

**C** 高于不支付股息股票的持有成本。
higher than the cost of carry on a stock with no dividends.

## 参考答案 / Supplied answer A

持有现货需要承担融资成本，但持有期间收到的股息会抵减净持有成本。因此在融资利率等其他条件相同时，支付股息股票的净持有成本较低，相应的无套利远期价格也较低。

Holding the stock incurs financing costs, while dividends received during the holding period offset the net cost of carry. With financing rates and other terms unchanged, a dividend-paying stock has a lower net cost of carry and a lower no-arbitrage forward price.

> With continuous yields: $F_0 = S_0 \times \exp[(r - q) \times T]$

易错点 / Take care

股息是持有现货的收益，不是额外费用；在远期定价公式中，股息收益率q从融资利率r中扣除。

Dividends are a benefit of holding the spot asset, not an additional expense. The dividend yield q is subtracted from the financing rate r.

# 原题 Q65

代表题 / CORE

无套利、远期价格与持有成本 | No arbitrage, forward prices and carry

其他条件不变，若无风险利率在远期合约签订后立即上升，该合约对远期买方的价值将如何变化?

All else being equal, if the risk-free rate increases immediately after the inception of a forward contract, the value of the contract to the forward buyer will:

**A** 下降。
decrease.

**B** 保持不变。
stay the same.

**C** 上升。
increase.

## 参考答案 / Supplied answer C

签约后，旧合约的交割价K保持不变。标准无收益情形下，远期多头的价值为当前现货价格减去旧交割价的现值。其他条件不变时，利率上升使PV(K)下降，因此现有远期多头价值上升。

After inception, the existing contract's delivery price K remains fixed. In the standard no-income case, the long forward's value is today's spot price minus the present value of that fixed delivery price. Holding other inputs constant, a higher interest rate reduces PV(K) and increases the long forward's value.

$V_{long,t} = S_t - K / (1 + r)^{(T-t)}$

K and $S_t$ fixed: r increases -> PV(K) decreases -> $V_{long,t}$ increases

易错点 / Take care

不要把旧合约交割价K随着市场重新设定。新签远期的公平交割价会上升；这正是旧远期多头更有价值的原因。

Do not reset the old contract's delivery price K. The fair delivery price on a new forward rises, which is why the existing long forward becomes more valuable.

# 期货结算与远期期货差异

Futures settlement and forward-futures differences

## 每日结算：把浮动盈亏变成现金流

### Daily settlement turns gains and losses into cash flows

期货价格会变。每日按新结算价计算盈亏，通过变动保证金收付，随后合约价值重置至零。初始保证金是履约担保，不是购买期货的权利金。

The futures price changes. Gains and losses are settled through variation margin, resetting contract value to zero at the new settlement price. Initial margin is performance collateral, not an option-like purchase premium.

> Daily gain to a long = units × multiplier × ($F_t$ - Fprevious)

集中清算、保证金和每日结算通常降低对手信用暴露，因此 Q22 选 futures，Q36 选 variation margin。风险降低不代表不存在风险。

Central clearing, margin and daily settlement generally reduce counterparty exposure. Thus Q22 selects futures and Q36 selects variation margin. Reduced risk is not zero risk.

## 正相关时，为什么多头喜欢期货?

### Why does positive correlation favor a futures long?

| 情形 / Scenario | 每日结算的影响 / Daily-settlement effect |
|---|---|
| 价格涨、利率也涨<br>Price and rate rise | 多头收到现金，可按更高利率再投资<br>The long receives cash to reinvest at higher rates |
| 价格跌、利率也跌<br>Price and rate fall | 多头需补现金，但融资利率较低<br>The long funds losses at lower rates |

在标准比较条件下，正相关使期货多头现金流更有利，期货价格通常高于同期限远期价格；负相关时方向相反。差异取决于利率波动和相关性。利率确定时，两者通常相同。

Under standard assumptions, positive correlation favors the futures long and generally supports a higher futures than forward price; negative correlation reverses the effect. Both rate volatility and correlation matter. Deterministic rates generally remove the difference.

## Q62 的“凸性”必须按题目口径理解

### Read Q62’s convexity term in context

易错点 / Take care

原答案是 B：3个月 MRR 的 FRA。FRA 在期初折现结算，分母随利率变，金额对利率是非线性的；本题比较下，期限更长的 FRA 影响更大。利率期货按利率变化线性计损益。市场上的 futures-forward convexity adjustment 是相关但不同的比较。

The supplied answer is B: the 3-month FRA. Advance FRA settlement has a rate-dependent denominator, making the cash amount nonlinear; the longer accrual period increases this effect in the question’s comparison. Rate futures have linear quoted-rate P&L. A market futures-forward convexity adjustment is a related but distinct comparison.

# 原题 Q22

期货结算与远期期货差异 | Futures settlement and forward-futures differences

下列哪种衍生品的交易对手违约风险通常最低?

Counterparty default risk is most likely lowest for which of the following types of derivatives?

**A** 互换。
Swaps

**B** 期货。
Futures

**C** 远期。
Forwards

## 参考答案 / Supplied answer B

交易所期货通过集中清算、保证金及每日结算来限制未结清的信用敞口，因此在本题所比较的典型结构中，交易对手违约风险最低。普通双边场外远期和互换的风险更依赖对手方信用与抵押安排。

Exchange-traded futures use central clearing, margin, and daily settlement to limit outstanding credit exposure. Among the typical structures compared here, futures therefore have the lowest counterparty default risk. Bilateral OTC forwards and swaps depend more on counterparty credit quality and collateral arrangements.

易错点 / Take care

Lowest 不等于 zero；期货仍可能带来追加保证金和流动性压力。

Lowest does not mean zero. Futures can still create margin calls and liquidity pressure.

# 原题 Q36

期货结算与远期期货差异 | Futures settlement and forward-futures differences

关于期货合约，以下哪一种说法正确?

A futures contract's:

**A** 盯市损益要等到到期才结算。
mark-to-market is not settled until maturity.

**B** 价格在到期前保持不变。
price remains fixed until the contract matures.

**C** 变动保证金可以降低交易对手信用风险。
variation margin reduces counterparty credit risk.

## 参考答案 / Supplied answer C

期货通常每天盯市并以变动保证金结算盈亏，使未清偿的信用风险敞口不会一直积累到到期。期货报价会随市场变化，A、B 均与期货机制相反。保证金和清算制度降低风险，但不代表所有风险归零。

Futures are normally marked to market daily, with gains and losses settled through variation margin, preventing unpaid credit exposure from accumulating until maturity. Futures quotes change with market conditions, so A and B are incorrect. Margin and clearing reduce risk but do not eliminate every risk.

易错点 / Take care

期货已实现的每日现金流与期末总盈亏要区分；保证金不是期权权利金。

Separate daily realized cash flows from total terminal profit, and do not confuse margin with an option premium.

# 原题 Q39

代表题 / CORE

期货结算与远期期货差异 | Futures settlement and forward-futures differences

对于相同标的与到期日，当期货价格和利率呈哪一种关系时，期货多头比远期多头更有吸引力?

Long futures contracts are more attractive than long forward positions for the same underlying and maturity when futures prices and interest rates are:

**A** 负相关。
negatively correlated.

**B** 不相关。
uncorrelated.

**C** 正相关。
positively correlated.

## 参考答案 / Supplied answer C

正相关意味着期货上涨产生每日现金收益时，市场利率也往往较高，可以按更高利率再投资这些收益。期货下跌需要补付现金时，利率也往往较低，融资相对便宜。远期没有同样的逐日结算再投资优势，因此在这一条件下期货多头更有吸引力。

With positive correlation, futures gains tend to arrive when interest rates are high and can be reinvested at attractive rates. Losses tend to require funding when rates are lower. A forward lacks these daily settlement and reinvestment effects, making long futures more attractive under this condition.

易错点 / Take care

这里讨论的是每日结算现金流与利率的相关性，不是单纯预测标的会上涨。

The issue is the correlation between daily settlement cash flows and rates, not merely a forecast that the underlying will rise.

# 原题 Q40

期货结算与远期期货差异 | Futures settlement and forward-futures differences

远期价格与期货价格之间的差异由下列哪些因素决定?

The differential between forward and futures prices is determined by which of the following?

**A** 只有利率波动率。
Interest rate volatility only

**B** 只有期货价格与利率之间的相关性。
The correlation between futures prices and interest rates only

**C** 利率波动率以及期货价格与利率之间的相关性。
Both interest rate volatility and the correlation between futures prices and interest rates

## 参考答案 / Supplied answer C

期货逐日结算，而远期通常到期结算；利率的不确定性使这些不同时间的现金流具有不同再投资或融资价值。差异既取决于利率能波动多大，也取决于现金收益出现时利率处于高位还是低位，即价格与利率的相关性。在利率确定等标准条件下，二者价格相同。

Daily settlement of futures, compared with maturity settlement of forwards, creates reinvestment and funding effects when rates are uncertain. Their importance depends on the amount of rate volatility and on whether gains occur when rates are high or low, captured by price-rate correlation. Under standard conditions with deterministic rates, forward and futures prices coincide.

易错点 / Take care

仅记住正相关而忽略利率波动不足以完整解释价格差异。

Remembering the sign of correlation alone is insufficient; the amount of interest rate uncertainty also matters.

# 原题 Q62

代表题 / CORE

期货结算与远期期货差异 | Futures settlement and forward-futures differences

以下哪一种利率衍生品最可能具有最大的凸性偏差?

Which of the following interest rate derivatives most likely has the largest convexity bias?

**A** 以1个月市场参考利率为标的的远期利率协议。
Forward rate agreement on a 1-month market reference rate

**B** 以3个月市场参考利率为标的的远期利率协议。
Forward rate agreement on a 3-month market reference rate

**C** 以3个月市场参考利率为标的的利率期货合约。
Interest rate futures contract on a 3-month market reference rate

## 参考答案 / Supplied answer B

本题采用所给答案的“结算金额对利率的非线性”口径。FRA在计息期初结算时，须把期末利息差额按市场参考利率折现，其分母包含1 + L×tau，因而不是利率的线性函数。3个月FRA的折现期间长于1个月FRA，凸性效应更大；题中标准利率期货的每基点盈亏按固定金额线性计算，所以选B。

This question uses the supplied solution’s meaning of convexity: nonlinearity of the settlement amount in the reference rate. A start-of-period FRA settlement discounts the end-of-period interest difference using a denominator of 1 + L × tau. A three-month FRA has a longer discounting interval than a one-month FRA and therefore a larger convexity effect. The standard interest rate futures payoff considered here is linear per basis point, so B is the intended answer.

> FRA settlement = N × (L - K) × tau / (1 + L × tau)
>
> Longer accrual/discounting interval tau -> greater FRA settlement curvature

易错点 / Take care

不要把“FRA预结算金额的曲率”与市场实务中“期货利率和远期利率之间的凸性调整”混为一谈。本题按第一种口径比较，不能看到futures就选C。

Do not confuse curvature in a discounted FRA settlement with the market convexity adjustment between futures and forward rates. This question compares the first concept; the word futures does not make C correct.

审校说明 / Wording and scope

原答案明确为B。其“期货没有凸性偏差”应按本题线性每基点支付的上下文理解，不能泛化为所有期货和远期之间都无需凸性调整。

The supplied answer is explicitly B. Its statement that futures have no convexity bias refers to the linear per-basis-point payoff in this question; it is not a claim that futures-forward convexity adjustments never exist.

# 远期利率、FRA 与互换

Forward rates, FRAs and swaps

## 从零息债价格提取远期利率

### Extract a forward rate from zero-coupon bond prices

两条无风险投资路径应有相同终值：一直投资到长期到期日；或先投资到较早日期，再按隐含远期利率续投。若两只零息债面值相同，可直接用价格比。

Two riskless investment paths must reach the same terminal value: invest to the later maturity, or invest to the earlier maturity and reinvest at the implied forward rate. Equal-face-value zero-coupon bonds allow a direct price ratio.

> Annual-compounded forward rate: f(t,T) = [P(0,t)/P(0,T)]^(1/(T-t)) - 1
>
> Q6: f(2,3) = 96/93 - 1 = 3.2258% ≈ 3.23%

## FRA 多头 = 付固定、收浮动

### Long FRA = pay fixed, receive floating

K 为约定固定利率，L 为期初确定的市场参考利率，α 为计息年数。L 高于 K 时，多头受益；可用于对冲未来借款利率上升。

K is the agreed fixed rate, L the observed market reference rate, and α the accrual fraction. A long FRA gains when L exceeds K and can hedge higher future borrowing costs.

> End-of-period difference = N × (L - K) × α
>
> Advance settlement = N × (L - K) × α / (1 + L × α)

| 时点 / Time | 发生什么 / What happens |
|---|---|
| 订约 t0 / Contract date | 公允定价的标准 FRA 期初价值及净付款为零<br>Fair standard FRA: zero initial value and net payment |
| 计息起点 t1 / Accrual start | 观察 L，并按惯例支付折现后的差额<br>Observe L and settle the discounted difference |
| 计息终点 t2 / Accrual end | 未折现差额对应这一天；不要重复结算<br>Undiscounted interest difference belongs to this date |

易错点 / Take care

Q31：利率期货报价约为 100 - 利率（百分数）。利率上升，期货价格下降，所以 long FRA 与 short interest-rate futures 都获利。

Q31: a rate future is quoted approximately as 100 minus the rate in percent. Higher rates lower its price, benefiting both a long FRA and a short interest-rate future.

# 远期利率、FRA 与互换

Forward rates, FRAs and swaps

## 互换是一系列固定与浮动现金流的交换

### A swap exchanges a series of fixed and floating cash flows

标准同币种利率互换一般不交换名义本金，按各期末净额结算。固定利率通常贯穿所有期间；每期浮动利率按约定市场参考利率重设。

A standard single-currency interest rate swap generally does not exchange notional principal and settles net at period ends. The fixed rate usually stays constant while the floating rate resets.

> Fixed-rate payer receives: $N \times (L - K) \times \alpha$
>
> Fixed-rate receiver receives: $N \times (K - L) \times \alpha$

| 持仓 / Position | 利率上升 / Higher floating rates | 原题 / Q |
|---|---|---|
| 付固定、收浮动<br>Pay fixed / receive floating | 价值通常上升<br>Value generally increases | 31, 46 |
| 收固定、付浮动<br>Receive fixed / pay floating | 价值通常下降<br>Value generally decreases | 41 |

## Q46：先写“收到的减付出的”

### Q46: write received minus paid

> Year 3 expected L = 1.35%; fixed K = 1.95%; N = 10,000,000
>
> CF = 10,000,000 × (0.0135 - 0.0195) × 1 = -60,000

问的是第 3 年一次结算，不是前三年现金流的总和，也不是整份互换的现值。题目给出年付，所以 α=1。

The question asks for a single Year 3 settlement, not the sum of three settlements or the present value of the whole swap. Annual payments give α=1.

## 为什么一串 FRA 不等于每笔都与互换相同?

### Why is a strip of FRAs not identical in every cash flow?

标准公平定价的 FRA 与互换都可没有期初净付款，这是 Q29 的共同点。各 FRA 的固定利率通常等于各期远期利率，互换却使用同一个固定利率；FRA 与互换也可在不同时间结算。

Fair standard FRAs and a par swap can both have no upfront net payment: Q29’s common feature. Individual FRA fixed rates normally differ across periods, whereas a swap uses one fixed rate. Settlement timing can also differ.

易错点 / Take care

“一串远期的组合”是理解互换的方式，不代表每个标准 FRA 的固定利率和结算现金流都相同。

Viewing a swap as a collection of forward exposures does not make the fixed rates and settlement cash flows of every standard FRA identical.

# 原题 Q06

代表题 / CORE

远期利率、FRA 与互换 | Forward rates, FRAs and swaps

两年期和三年期政府基准零息债券的价格分别为 96 和 93（面值均为 100）。两年后开始、期限为一年的隐含远期利率最接近多少？

Two-year and three-year government benchmark zero-coupon bonds are priced at 96 and 93 (per 100 face value), respectively. The implied one-year forward rate in two years' time is closest to:

**A** 3.00%.

**B** 3.23%.

**C** 3.36%.

## 参考答案 / Supplied answer B

债券价格除以面值就是相应期限的折现因子，分别为 0.96 和 0.93。连接第 2 年与第 3 年的一年远期增长因子，等于两年折现因子除以三年折现因子，因此无需先计算两个零息利率。相除后减 1，得到 3.23%。

Divide each bond price by face value to obtain discount factors of 0.96 and 0.93. The one-year forward growth factor from Year 2 to Year 3 equals the two-year discount factor divided by the three-year discount factor. There is no need to calculate both zero rates first; subtracting one gives 3.23%.

DF(2) = 96/100 = 0.96; DF(3) = 93/100 = 0.93

1 + f_(2,3) = DF(2)/DF(3) = 0.96/0.93

f_(2,3) = 0.032258 = 3.23%

易错点 / Take care

远期利率来自增长因子的比值，不是两个即期利率或债券价格的简单差。

The forward rate comes from a ratio of growth factors, not a simple difference between spot rates or bond prices.

# 原题 Q29

远期利率、FRA 与互换 | Forward rates, FRAs and swaps

一系列覆盖相同期间、采用相同市场参考利率的远期利率协议，与一份利率互换最有可能具有相同的哪一项?

A series of forward rate agreements and an interest rate swap contract covering the same periods and using the same market reference rate will most likely have the same:

**A** 固定利率。
fixed rates.

**B** 合约起始时的现金流。
cash flows upfront.

**C** 结算现金流。
settlement cash flows.

## 参考答案 / Supplied answer B

标准、按市场公允利率订立的 FRA 和利率互换都具有零初始价值，通常不交换期初付款。不同期间的 FRA 固定利率通常不同，而标准互换在整个合约中使用同一固定利率。传统 FRA 通常在计息期初折现结算，互换通常在各计息期末结算，因此结算现金流也未必相同。

Standard FRAs and interest rate swaps entered at fair market rates have zero initial value and normally no upfront payment. FRAs for different periods generally have different fixed rates, while a standard swap uses one fixed rate throughout. Traditional FRAs settle on a discounted basis at the start of the interest period, whereas swaps normally settle at each period-end.

易错点 / Take care

swap 可理解为一组远期现金流，但不等于每一份独立按市价定价的 FRA 都有相同固定利率。

A swap resembles a package of forward cash flows, but independently priced at-market FRAs need not share its fixed rate.

审校说明 / Wording and scope

这里的无期初付款指标准、公允定价合约，不把抵押品或保证金要求视作合约价格。

No upfront payment refers to standard at-market contracts; collateral or margin is not treated as the contract price.

# 原题 Q31

代表题 / CORE

远期利率、FRA 与互换 | Forward rates, FRAs and swaps

当市场参考利率升至初始固定利率以上时，下列哪些衍生品头寸会获利?

Which of the following derivatives realize a gain as the market reference rate rises above the initial fixed rate?

**A** 只有远期利率协议多头。
Long forward rate agreements only

**B** 只有利率期货空头。
Short interest rate futures contracts only

**C** 远期利率协议多头与利率期货空头都会获利。
Both long forward rate agreements and short interest rate futures contracts

## 参考答案 / Supplied answer C

本题中的 FRA 多头是支付固定利率、收取浮动利率的一方；参考利率上升使其净收款增加。标准短期利率期货通常按 100 减去利率报价，利率升高使期货报价下降，因此期货空头获利。两个头寸对利率上升的受益方向一致。

The long FRA in this convention pays fixed and receives floating, so a higher reference rate increases its net receipt. Standard short-term interest rate futures are commonly quoted as 100 minus the rate; a higher rate therefore lowers the futures quote and benefits the short. Both positions gain from rising rates.

> Long FRA settlement sign: L - K.
>
> Short-term rate futures quote = 100 - rate (in percent).

易错点 / Take care

看见 long 就认为一定受益于利率上升会出错；先判断合约的报价方式。

Do not assume every long position benefits from a rise in interest rates; first identify the contract's quoting convention.

# 原题 Q33

远期利率、FRA 与互换 | Forward rates, FRAs and swaps

以下哪一种说法最准确？一份标准利率互换具有哪项特征？

Which of the following statements is most accurate? A standard interest rate swap has:

**A** 对称的收益结构。
a symmetric payoff profile.

**B** 在合约开始时交换本金现金流。
the principal cash flow exchanged upfront.

**C** 各期结算在计息期初进行。
periodic settlements that occur at the beginning of each period.

## 参考答案 / Supplied answer A

标准利率互换属于远期承诺，双方都必须履约，利率向有利或不利方向变动会带来相应的收益或损失，因此属于对称收益结构。普通单一货币互换只以名义本金计算利息，通常不交换本金。定期利息通常在各计息期末结算。

A standard interest rate swap is a forward commitment: both parties must perform and favorable or unfavorable rate movements create corresponding gains or losses, giving a symmetric payoff profile. A plain single-currency swap uses notional principal to calculate interest and normally does not exchange principal. Interest is generally settled at each interest period-end.

易错点 / Take care

不要把单一货币利率互换与可能交换本金的货币互换混为一谈。

Do not confuse a single-currency interest rate swap with a currency swap that may exchange principal.

# 原题 Q41

远期利率、FRA 与互换 | Forward rates, FRAs and swaps

从固定利率收取方的角度看，如果市场参考利率上升，利率互换的价值会如何变化?

From the fixed-rate receiver's perspective, if the market reference rate increases, the value of a swap contract:

**A** 下降。
decreases.

**B** 保持不变。
stays the same.

**C** 上升。
Increases.

## 参考答案 / Supplied answer A

固定利率收取方收固定、付浮动。参考利率上升使其要支付的浮动利息增加，而收到的固定利息不变，所以原有互换变得较不利，价值下降。反过来，固定利率支付方通常受益于利率上升。

The fixed-rate receiver receives fixed and pays floating. A higher reference rate increases the floating payment while leaving the fixed receipt unchanged, making the existing swap less favorable and lowering its value. The fixed-rate payer generally benefits from rising rates instead.

> Receiver net interest = N × (K-L) × accrual fraction.

易错点 / Take care

先写清楚 receive fixed = pay floating，再判断利率变动方向。

Write receive fixed = pay floating before deciding how a rate move affects the position.

# 原题 Q46

代表题 / CORE

远期利率、FRA 与互换 | Forward rates, FRAs and swaps

一份名义本金为1,000万美元、每年结算一次的利率互换，固定互换利率为1.95%。隐含远期利率如下。预计固定利率支付方第3年的定期结算金额最接近多少?

A $10 million interest rate swap with annual payments has a fixed swap rate of 1.95%. The implied forward rates are shown below. The periodic settlement value in Year 3 for the fixed-rate payer is expected to be closest to:

| 年份 / Year | 隐含远期利率 / Implied forward rate |
|---|---|
| 1 | 0.50% |
| 2 | 1.15% |
| 3 | 1.35% |

**A** -$95,000.

**B** -$60,000.

**C** $60,000.

## 参考答案 / Supplied answer B

固定利率支付方支付固定利息，同时收取浮动利息。因此其净结算金额为“浮动减固定”，乘以名义本金和计息期间。第3年浮动利率预计为1.35%，低于1.95%，所以应净支付60,000美元。

The fixed-rate payer pays fixed interest and receives floating interest. Its net settlement is therefore floating minus fixed, multiplied by notional principal and the accrual fraction. The expected Year 3 floating rate is 1.35%, below 1.95%, so the payer makes a net payment of $60,000.

> Settlement = N × (L - K) × tau
>
> Settlement = 10,000,000 × (0.0135 - 0.0195) × 1 = -60,000

易错点 / Take care

结算金额不等于互换在今天的价值；本题只问第3期现金流，不需要折现，也不交换1,000万美元本金。

A periodic settlement is not the swap value today. This question asks only for the Year 3 cash flow: do not discount it or exchange the $10 million notional.

# 期权四种头寸、价值与利润

Option positions, payoff and profit

## 先看方向，再算行权价值，最后计入权利金

**Position first, exercise value second, premium last**

K 是执行价，ST 是到期标的价，c0/p0 是期初权利金。以下利润按本套题口径，不计权利金融资成本和交易费；若题目给合约数量或乘数，再乘上去。

K is the strike, ST the terminal underlying price, and c0/p0 the initial premium. Profit below follows this question set: no premium financing cost or fees. Apply any specified units or contract multiplier.

| 头寸 / Position | 到期 payoff | 净利润 / Profit |
|---|---|---|
| 买入 Call / Long call | max(ST-K, 0) | max(ST-K, 0) - c0 |
| 卖出 Call / Short call | -max(ST-K, 0) | c0 - max(ST-K, 0) |
| 买入 Put / Long put | max(K-ST, 0) | max(K-ST, 0) - p0 |
| 卖出 Put / Short put | -max(K-ST, 0) | p0 - max(K-ST, 0) |

> Call break-even: $S_T = K + c_0$
>
> Put break-even: $S_T = K - p_0$

## Q4：同一笔交易，两个答案都可能出现在选项里

**Q4: two different quantities can both appear as choices**

> $K=25$; $c_0=5$; $S_T=30$
>
> Short-call payoff = -max(30-25, 0) = -5
>
> Short-call profit = 5 - 5 = 0

题目问到期 value to the call seller，答案为 -5；如果问 profit，才是 0。Q16 的买入 call 到期虚值，payoff 为 0，但买方已亏掉 4 的权利金。

The question asks for the seller’s terminal value, so the answer is -5. Profit would be 0. In Q16, an out-of-the-money call pays 0, but its buyer has lost the premium of 4.

易错点 / Take care

绝不默认乘 100。Q56 明确一个指数点值 1 美元；Q55 则明确有 2,000 股。

Never assume a multiplier of 100. Q56 specifies USD 1 per index point, while Q55 explicitly specifies 2,000 shares.

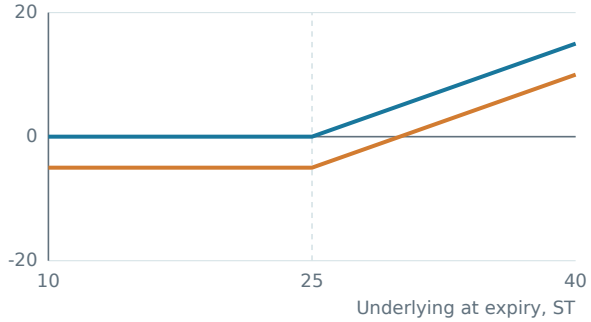
# 期权四种头寸、价值与利润

Option positions, payoff and profit

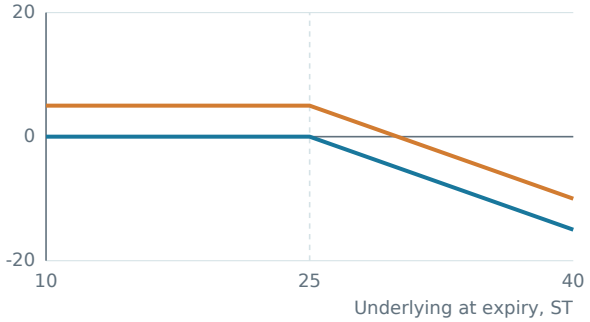
## 用图看懂买方与卖方的镜像关系

**Long and short positions are mirror images**

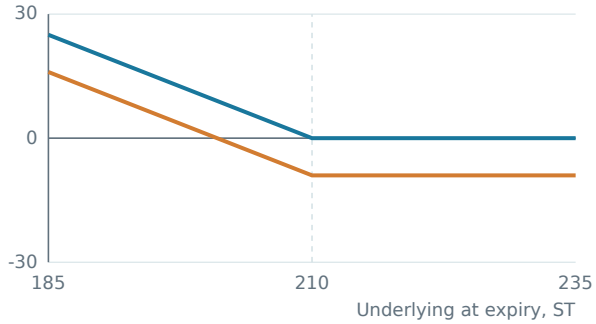
**Long call / Q4**



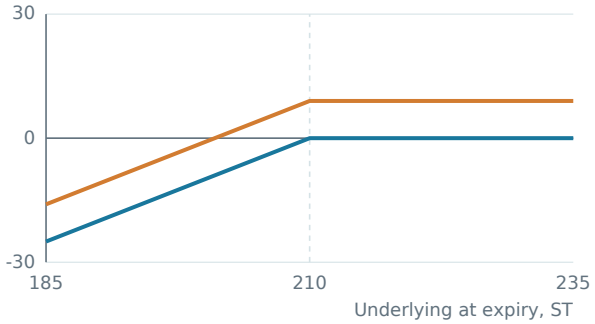

**Short call / Q4**




**Long put / Q61**




**Short put / Q61**




图中蓝线是到期 payoff，橙线是包含权利金的 profit。Call 采用 Q4 的 K=25、权利金=5；Put 采用 Q61 的 K=210、权利金=9。

Blue curves show terminal payoff; orange curves include the premium and show profit. Call parameters come from Q4 (K=25, premium=5); put parameters come from Q61 (K=210, premium=9).

| 关系 / Condition | Call | Put |
|---|---|---|
| 实值 / In the money | S > K | S < K |
| 平值 / At the money | S = K | S = K |
| 虚值 / Out of the money | S < K | S > K |

内在价值取决于“现在立刻行权”能拿多少；时间价值 = 期权价格 - 内在价值。Q26 的欧式 put 在到期前虚值，内在价值为零，但在正常波动假设下仍可有正时间价值。

Intrinsic value measures immediate exercise value. Time value equals option price minus intrinsic value. The out-of-the-money European put in Q26 has zero intrinsic value but can still have positive time value under ordinary volatility assumptions.

易错点 / Take care

Q57 给的是到期 call value=17.80，所以 ST=130+17.80。Q59 选 A 的考点是两条腿都因上涨受益；不能据此说 long forward + long put 的合计价值绝不会上涨。

Q57 gives call value at maturity, so ST=130+17.80. Q59 selects A because both legs benefit from a rise; it does not establish that a long forward plus a long put can never rise in total value.

题源 / Source questions: Q4, Q16, Q23, Q26, Q48, Q56, Q57, Q58, Q59, Q61

# 原题 Q04

代表题 / CORE

期权四种头寸、价值与利润 | Option positions, payoff and profit

投资者取得某看涨期权的下列资料。到期时，若标的资产价格为 $30，则该看涨期权对卖方的到期价值是多少?

An investor gathers the following information about a call option. At expiration, if the price of the underlying is $30, the value of the call option to the call seller is:

| 项目 Item | 数值 Value |
|---|---|
| 期权权利金 Option premium | $5 |
| 执行价 Exercise price | $25 |
| 期初标的价格 Underlying price at initiation | $15 |

**A** -$5.

**B** $0.

**C** $10.

## 参考答案 / Supplied answer A

先计算看涨期权买方的到期价值：股价高于执行价 $5，因此买方取得 $5。卖方承担相反的现金流，故到期价值为 -$5。题目问 value，而非包含最初收到权利金的净利润；若问净利润，才是 $5 - $5 = $0。

The call buyer receives $5 at expiration because the stock price exceeds the strike by $5. The seller has the opposite payoff, or -$5. The question asks for value at expiration, not profit after including the initial premium; the latter would be $5 minus $5, or zero.

Call payoff = $\max(S_T - K, 0) = \max(30 - 25, 0) = \$5$

Short-call value at expiry = $-\max(S_T - K, 0) = -\$5$

Short-call profit = $c_0 - \max(S_T - K, 0) = 5 - 5 = \$0$

易错点 / Take care

Value / payoff 不计初始权利金；profit 才把权利金收支计入。

Value or payoff excludes the initial premium; profit includes the premium received or paid.

# 原题 Q16

期权四种头寸、价值与利润 | Option positions, payoff and profit

投资者以 $4 买入一份执行价为 $27 的看涨期权。到期时股价为 $22，则该期权的到期收益为哪一种情况？

An investor buys a call option for $4 that has an exercise price of $27. At expiration, if the stock price is $22, the call option payoff is:

**A** 负值。
negative.

**B** 零。
zero.

**C** 正值。
positive.

## 参考答案 / Supplied answer B

股价低于执行价，按 $27 买入市价仅为 $22 的股票没有吸引力，因此期权不行权，到期收益为零。已经支付的 $4 是期权成本，会使净利润为 -$4，但不会使买方的期权到期收益变成负值。

The stock is below the strike, so paying $27 for a stock worth only $22 is unattractive. The option expires unexercised and its payoff is zero. The $4 premium makes the investor's profit -$4, but it does not make the long option's payoff negative.

> Call payoff = max(22 - 27, 0) = $0
>
> Long-call profit = 0 - 4 = -$4

易错点 / Take care

买入期权的 payoff 最低为零；profit 的最低值通常是损失全部权利金。

A long option has a minimum payoff of zero; its minimum profit normally equals the premium paid with a negative sign.

# 原题 Q23

期权四种头寸、价值与利润 | Option positions, payoff and profit

其他条件不变，看涨期权的行权价格较低时，该期权到期处于价内状态的可能性会如何变化?

When the strike price of a call option is lower, the likelihood of the option expiring in-the-money is:

**A** 更低。
lower

**B** 不变。
unchanged.

**C** 更高。
higher.

## 参考答案 / Supplied answer C

看涨期权价内的条件是到期标的价格 S_T 高于行权价 X。降低 X 后，更多可能的 S_T 满足这个条件，因此价内概率提高。较低的行权价也使相同 S_T 下的行权收益更高。

A call finishes in the money when S_T exceeds X. Lowering X makes this condition true for more possible terminal prices, so the probability of finishing in the money rises. It also increases the exercise payoff for a given terminal price.

> Call is in the money when $S_T > X$.

易错点 / Take care

价内与最终盈利不同；看涨期权买方的盈亏平衡价还要加上权利金。

In the money does not necessarily mean profitable: the call buyer must also recover the premium.

# 原题 Q26

期权四种头寸、价值与利润 | Option positions, payoff and profit

到期前，如果标的价格高于行权价，一份欧式看跌期权的哪一项价值为正?

Before expiration, if the price of the underlying is above the exercise price, the European put option has a positive:

**A** 时间价值。
time value.

**B** 内在价值。
intrinsic value.

**C** 行权价值。
exercise value.

## 参考答案 / Supplied answer A

看跌期权的行权价值为 max(X-S,0)；当 S>X 时，它处于价外状态，行权价值等于零。题目采用通常仍有不确定性的期权情境，未来价格仍可能跌到 X 以下，所以期权可以具有正的时间价值。内在价值与行权价值在这里是同义词。

A put's exercise value is max(X-S,0), so it is zero when S exceeds X. Under the usual uncertain-price setting assumed in the question, the underlying could still fall below X before expiry, giving the option positive time value. Intrinsic value and exercise value mean the same thing here.

> Intrinsic value = max(X-S, 0) = 0.
>
> Time value = Put premium - Intrinsic value.

易错点 / Take care

欧式期权不能在到期前行权，但仍可讨论其内在价值及时间价值。

Although a European option cannot be exercised before expiry, its intrinsic and time values can still be defined.

# 原题 Q48

期权四种头寸、价值与利润 | Option positions, payoff and profit

一份看跌期权的价格为2美元，行权价为45美元。若标的资产到期价格为41美元，看跌期权买方的利润是多少?

Consider a put option selling for $2 in which the exercise price is $45. What is the profit for a put buyer if the price of the underlying at expiration is $41?

**A** -$2

**B** $2

**C** $4

## 参考答案 / Supplied answer B

看跌期权在标的价格低于行权价时产生支付。本题到期支付为45减41，等于4美元。买方开始时已支付2美元权利金，因此利润为2美元。

The put pays when the underlying finishes below the strike. Its expiration payoff is $45 minus $41, or $4. The buyer paid a $2 premium, so profit is $2.

> Put payoff = $\max(K - S_T, 0) = \max(45 - 41, 0) = 4$
>
> Long put profit = payoff - premium = 4 - 2 = 2

易错点 / Take care

4美元是支付，不是利润；求profit必须减去已付权利金。

$4 is the payoff, not the profit. Subtract the premium whenever the question asks for profit.

# 原题 Q56

代表题 / CORE

期权四种头寸、价值与利润 | Option positions, payoff and profit

交易员以25美元卖出一份股指看涨期权，行权价为2,400点。指数每变动1点对应1美元。到期时指数为2,450点，该交易员的利润是多少?

A trader sells a call option on a stock index with a strike price of 2,400 for $25. The value of a one-point move in the index is $1. At expiration, the stock index is trading at 2,450. The trader's profit is:

**A** -$50.

**B** -$25.

**C** $25.

## 参考答案 / Supplied answer B

指数高于行权价50点，看涨期权卖方需支付50美元。卖方最初收取25美元权利金，所以总利润为25减50，等于亏损25美元。

The index finishes 50 points above the strike, so the call seller owes $50. The seller originally received a $25 premium, giving a total profit of $25 - $50 = -$25.

> Call payoff to buyer = max(2,450 - 2,400, 0) × $1 = $50
>
> Short call profit = premium - payoff = 25 - 50 = -25

易错点 / Take care

先算买方支付，再取相反数得到卖方支付；最后加上卖方收到的权利金，并核对每点乘数。

First calculate the buyer’s payoff, then reverse its sign for the seller. Add the premium received and check the dollar multiplier per index point.

# 原题 Q57

期权四种头寸、价值与利润 | Option positions, payoff and profit

投资者以5.75美元买入一份行权价为130美元的看涨期权。若该期权到期价值为17.80美元，标的资产到期价格最接近多少?

An investor buys a call for $5.75 that has a strike price of $130. If the value at expiration for this call is $17.80, the price of the underlying at expiration is closest to:

**A** $112.20.

**B** $142.05.

**C** $147.80.

## 参考答案 / Supplied answer C

看涨期权到期价值大于0，表示其处于价内。到期价值等于S_T - K，因此标的资产价格为130 + 17.80 = 147.80美元。题目给出的是期权价值，不是投资者净利润，不能把5.75美元权利金混入这一步。

A positive call value at expiration means the option is in the money. Its value equals S_T - K, so the underlying price is $130 + $17.80 = $147.80. The question gives option value rather than the investor's net profit, so the $5.75 premium does not enter this calculation.

$$17.80 = S_T - 130$$

$$S_T = 130 + 17.80 = 147.80$$

易错点 / Take care

value at expiration是支付，不是profit。若另问投资者利润，才计算17.80 - 5.75 = 12.05美元。

Value at expiration is the payoff, not profit. Only if investor profit were requested would you compute $17.80 - $5.75 = $12.05.

# 原题 Q58

期权四种头寸、价值与利润 | Option positions, payoff and profit

投资者卖出一份具有下表条款的欧式看跌期权。若标的资产到期价格为1,340，卖方的利润是多少?

An investor sells a European put option with the following characteristics. If the price of the underlying at expiration is 1,340, the profit for the seller is:

| 项目 / Item | 数值 / Value |
|---|---|
| 看跌期权价格 / Put price | 30 |
| 行权价 / Exercise price | 1,320 |

**A** 10.

**B** 20.

**C** 30.

## 参考答案 / Supplied answer C

标的资产到期价格1,340高于行权价1,320，看跌期权为价外，买方不会行权。卖方无需支付行权差额，完整保留最初收取的30权利金，因此利润为30。

The underlying finishes at 1,340, above the 1,320 strike. The put is out of the money and expires unexercised. The seller pays no exercise amount and retains the entire premium of 30, so profit is 30.

> Put payoff = max(1,320 - 1,340, 0) = 0
>
> Short put profit = 30 - 0 = 30

易错点 / Take care

期权支付不能为负；1,320 - 1,340 = -20必须经max(・,0)截断为0。

An option holder's exercise payoff cannot be negative. The -20 difference must be floored at zero by max(·, 0).

# 原题 Q59

期权四种头寸、价值与利润 | Option positions, payoff and profit

某分析师收集了下列三个组合的信息，每个组合都包含以同一资产为标的的两种衍生品。其他条件不变，哪一个组合将受益于标的资产价格上涨?

An analyst gathers the following information about three portfolios each consisting of two derivatives on the same underlying. All else being equal, which portfolio will benefit from an increase in price of the underlying?

| 组合 / Portfolio | 远期头寸 / Forward position | 期权头寸 / Option position |
|---|---|---|
| 1 | 多头 / Long | 买入看涨 / Long call |
| 2 | 多头 / Long | 买入看跌 / Long put |
| 3 | 空头 / Short | 买入看涨 / Long call |

**A** 组合1。
Portfolio 1

**B** 组合2。
Portfolio 2

**C** 组合3。
Portfolio 3

## 参考答案 / Supplied answer A

题目意在识别两条腿都具有看涨方向的组合。标的价格上涨有利于远期多头，也有利于看涨期权多头，因此组合1最明确地同时从两条腿受益。组合2和3则包含价格敏感方向相反的两条腿。

The intended test is to identify the portfolio with two bullish legs. A rise in the underlying benefits both a long forward and a long call, so Portfolio 1 unambiguously benefits through both positions. Portfolios 2 and 3 combine legs with opposing price sensitivities.

### 易错点 / Take care

不要把“两个头寸方向相反”误解成“组合总价值一定不受益”；总敏感度还取决于数量与各腿Delta。

Opposing individual sensitivities do not imply that the combined portfolio can never benefit. Net sensitivity also depends on position sizes and the legs’ deltas.

### 审校说明 / Wording and scope

原答案选A的理由是两条腿同向。题干若被理解为“整个组合只要净受益”，表述并不充分：例如一单位远期多头加一单位看跌期权，在标准条件下总Delta通常仍非负。讲义按原题意保留A，并明确这一边界。

The supplied key selects A because both legs are bullish. If “benefit” means only a net portfolio gain, the wording is incomplete: a unit long forward plus a unit long put generally still has nonnegative net delta under standard assumptions. Retain the intended answer A and state this limitation.

# 原题 Q61

代表题 / CORE

期权四种头寸、价值与利润 | Option positions, payoff and profit

投资者收集了下表中的看跌期权信息。若股票到期价格为200美元，买入该看跌期权的利润是多少?

An investor collects the following information about a put option. At expiration, if the price of the stock is $200, the investor's profit from buying the put is:

| 项目 / Item | 数值 / Value |
|---|---|
| 签约时股票价格 / Stock price at initiation | $220 |
| 行权价 / Strike price | $210 |
| 期权权利金 / Option premium | $9 |

**A** -$19.

**B** -$9.

**C** $1.

## 参考答案 / Supplied answer C

到期时股价200美元低于行权价210美元，看跌期权支付10美元。扣除买方支付的9美元权利金，利润为1美元。最初股价220美元不影响到期支付的计算。

At expiration, the $200 stock price is below the $210 strike, so the put pays $10. Subtract the $9 premium paid to obtain a $1 profit. The initial $220 stock price does not enter the expiration payoff calculation.

> Put payoff = max(210 - 200, 0) = 10
>
> Long put profit = 10 - 9 = 1

易错点 / Take care

以行权价和到期价格之差计算支付，不是以初始股价和到期股价之差计算。

Calculate the payoff from the strike and expiration price, not from the difference between the initial and final stock prices.

# 期权价值方向与上下界

Option value drivers and bounds

## 把影响方向记成成对关系

**Learn the directions in pairs**

| 其他不变，因素上升 / Increase | Call value | Put value |
|---|---|---|
| 标的价格 S / Underlying price | 上升 / Up | 下降 / Down |
| 执行价 K / Strike | 下降 / Down | 上升 / Up |
| 波动率 σ / Volatility | 上升 / Up | 上升 / Up |
| 无风险利率 r / Risk-free rate | 上升 / Up | 下降 / Down |
| 现金股息收益 / Dividend income | 下降 / Down | 上升 / Up |
| 剩余期限 T / Time remaining | 通常上升* / Usually up* | 不一定；通常上升*<br>Not unconditional* |

直觉：Call 喜欢买入价 K 低、标的 S 高；Put 喜欢卖出价 K 高、标的 S 低。买方的下方损失受权利金限制，较高波动通常增加有利尾部的价值。

A call likes a low purchase strike and a high asset price; a put likes a high sale strike and a low asset price. Limited downside for holders makes additional favorable tail outcomes valuable.

## 欧式、无分红的无套利边界

**European no-dividend no-arbitrage bounds**

$$PV(K) = K / (1 + r)^T$$

$$\max[0, S_0 - PV(K)] \le c_0 \le S_0$$

$$\max[0, PV(K) - S_0] \le p_0 \le PV(K)$$

Call 的价格不能高于直接买标的的价格，因此 Q34、Q63 分别选 B、A。Q11 的 max[PV(K)-S0,0] 是 put 下界。非负利率下 p≤K 也成立，但比欧式的 p≤PV(K) 更宽松。

A call cannot cost more than buying the underlying, giving Q34=B and Q63=A. Q11's expression is the put lower bound. With nonnegative rates, p≤K is also valid but weaker than the European bound p≤PV(K).

易错点 / Take care

*期限关系有条件：无分红、非负利率下，欧式 call 随期限不减；欧式 put 的期限关系可正可负。Q2 原解也承认这一点，只在 most likely 的一般情形下选期限较长、K 较高的 Option 3。

*Maturity effects require care: a no-dividend European call is nondecreasing with maturity at nonnegative rates; a European put need not be. Q2's supplied solution acknowledges this and selects Option 3 under its most-likely convention.

# 原题 Q01

期权价值方向与上下界 | Option value drivers and bounds

在其他条件相同的情况下，下列哪项最恰当地描述欧式看涨期权价值的关系？

All else held equal, the value of a European call option is best characterized as having a:

**A** 与标的资产价格负相关。
negative relationship with the price of the underlying.

**B** 与标的资产波动率负相关。
negative relationship with the volatility of the underlying.

**C** 与到期剩余时间正相关。
positive relationship with the time to expiration.

## 参考答案 / Supplied answer C

看涨期权赋予以固定价格买入的权利：标的价格越高，这项权利越有价值；波动率提高通常也使期权更有价值。因此 A、B 的方向都相反，本题选 C，采用期限越长通常越有价值的教材判断。

A call gives its holder the right to buy at a fixed price. A higher underlying price and greater volatility both increase its value under the standard model, so A and B have the wrong signs. C is the intended answer under the usual assumptions behind the maturity comparison.

易错点 / Take care

不要把期限关系背成适用于所有欧式期权的无条件定理。

Do not treat the maturity relationship as an unconditional rule for every European option.

审校说明 / Wording and scope

原答案选 C。无股息、非负利率等常用条件下可采用这一判断；离散股息等因素可能改变欧式看涨期权的期限比较。

The supplied answer is C. The usual result applies under assumptions such as no dividends and non-negative interest rates; discrete dividends can change a European call maturity comparison.

# 原题 Q02

代表题 / CORE

期权价值方向与上下界 | Option value drivers and bounds

同一标的资产的三个欧式看跌期权如下。在其他条件相同的情况下，哪一个的价值最可能最高?

All else being equal, which of the following European put options on the same underlying most likely has the highest value?

| 期权 Option | 到期剩余时间 Time to expiration | 执行价 Exercise price |
|---|---|---|
| 1 | 2 个月 / 2 months | $52 |
| 2 | 4 个月 / 4 months | $52 |
| 3 | 4 个月 / 4 months | $58 |

**A** 期权 1。
Option 1

**B** 期权 2。
Option 2

**C** 期权 3。
Option 3

## 参考答案 / Supplied answer C

先比较相同期限：期权 3 与期权 2 都在 4 个月后到期，而期权 3 的执行价更高，故其看跌权利更有价值。再按题目 most likely 的一般情形，4 个月的期权 2 比 2 个月的期权 1 更可能昂贵，因此选期权 3。

First compare the options with the same maturity: Options 2 and 3 both expire in four months, but Option 3 has the higher strike, making its put more valuable. Under the typical case implied by most likely, Option 2 is also more likely to exceed the shorter-dated Option 1. Therefore, Option 3 is the intended answer.

易错点 / Take care

Put 的执行价越高越贵；但欧式 put 的期限越长并不保证价值越高。

A higher strike increases put value, but a longer maturity does not always increase the value of a European put.

审校说明 / Wording and scope

原答案已明确：欧式看跌期权与期限可以正相关或负相关，正向关系较常见。本题是“最可能”的比较，不能改写为确定性排序定理。

The supplied explanation explicitly allows either a positive or a negative maturity effect for European puts, with a positive effect more common. This is a most-likely comparison, not a universal ordering theorem.

# 原题 Q03

期权价值方向与上下界 | Option value drivers and bounds

欧式看跌期权的价值与下列哪项正相关?

The value of a European put is directly related to the:

**A** 无风险利率。
risk-free rate.

**B** 执行价。
exercise price.

**C** 标的资产价值。
value of the underlying.

## 参考答案 / Supplied answer B

看跌期权允许按执行价卖出标的，所以执行价越高，卖出的条件越有利，期权价值越大。在标准比较中，标的价格上涨会降低看跌期权价值；利率上升降低执行价的现值，也通常降低看跌期权价值。

A put permits sale of the underlying at the strike, so a higher strike makes that right more valuable. Under the standard comparison, a higher underlying price reduces put value. Higher interest rates reduce the present value of the strike and generally reduce put value as well.

易错点 / Take care

记忆方向：put 喜欢高执行价、低股价；不要把 call 与 put 对利率的方向记反。

Remember: puts benefit from a higher strike and a lower stock price; their interest-rate direction differs from calls.

# 原题 Q11

代表题 / CORE

期权价值方向与上下界 | Option value drivers and bounds

下列哪一项等于“零”与“执行价现值减去现货价格”两者中的较大值?

Which of the following is equal to the greater of zero or the present value of the exercise price minus the spot price?

**A** 看跌期权价格的下界。
The lower bound of a put option

**B** 看涨期权价格的下界。
The lower bound of a call option

**C** 看跌期权价格的上界。
The upper bound of a put option

## 参考答案 / Supplied answer A

欧式看跌期权在无股息标的的常用无套利框架中，下界是 max[PV(K) - S_0, 0]。看跌期权提供未来按 K 卖出的权利，所以先考虑执行价现值与现货价之差，同时期权价格不能为负。看涨期权的下界则把两项的顺序反过来。

For a European put on a non-dividend-paying underlying in the usual no-arbitrage framework, the lower bound is max[PV(K) - S_0, 0]. The put grants a right to sell for K in the future, and its price cannot be negative. The call lower bound reverses the order of the stock price and discounted strike.

European put lower bound = $\max(PV(K) - S_0, 0)$

European call lower bound = $\max(S_0 - PV(K), 0)$

易错点 / Take care

到期前的价格下界用 PV(K)；立即行权价值用 K。不要把两者混同。

A pre-expiration price bound uses PV(K); intrinsic value uses K. They are different quantities.

# 原题 Q34

期权价值方向与上下界 | Option value drivers and bounds

看涨期权价格的上限是什么?

The upper bound of a call price is the:

**A** 行权价。
exercise price.

**B** 标的资产价格。
price of the underlying.

**C** 标的资产价格减行权价的现值。
underlying's price minus the present value of the exercise price.

## 参考答案 / Supplied answer B

看涨期权只是按 X 买入资产的权利，价值不应超过直接拥有该资产的价值。若看涨期权比标的本身还贵，投资者可卖出看涨期权并买入标的来构建套利。对于无分红欧式看涨期权，max(S0-PV(X),0) 是下限而非上限。

A call is only the right to buy the asset for X, so it cannot be worth more than owning the asset outright. If the call exceeded the underlying price, selling the call and buying the underlying would permit arbitrage. For a European call on a non-dividend-paying asset, max(S0-PV(X),0) is the lower bound, not the upper bound.

> Call upper bound: $c <= S_0$.

易错点 / Take care

认清 upper bound 与 lower bound；S0-PV(X) 不是看涨期权的上限。

Distinguish the upper from the lower bound: S0-PV(X) is not the call's upper bound.

# 原题 Q63

期权价值方向与上下界 | Option value drivers and bounds

看涨期权价值的上限是什么?

The upper bound of a call value is the:

**A** 标的资产价格。
underlying's price.

**B** 标的资产价格加上其行权价现值所得的金额，与0两者中的较大者。
underlying's price plus the present value of its exercise price or zero, whichever is greater.

**C** 标的资产价格减去其行权价现值所得的金额，与0两者中的较大者。
underlying's price minus the present value of its exercise price or zero, whichever is greater.

## 参考答案 / Supplied answer A

在本题标准条件下，看涨期权只提供以K买入标的的权利，而直接持有标的提供更完整的资产敞口。因此看涨期权不应比标的本身更贵，其上限为S。标的价格减去行权价现值、再与0比较，属于欧式看涨期权的下限结构。

Under the standard assumptions of this question, a call gives only the right to buy the underlying for K, while owning the asset provides the full underlying exposure. A call therefore cannot cost more than the asset itself; its upper bound is S. The maximum of zero and S minus the present value of K is a European call lower-bound expression.

> Call upper bound: $c <= S_0$
>
> European call lower bound: $c >= \max[S_0 - PV(K), 0]$

易错点 / Take care

上限是S；max[S - PV(K), 0]是下限，不能把上限与下限对调。

The upper bound is S; max[S - PV(K), 0] is a lower bound. Do not confuse upper and lower bounds.

# 看涨看跌平价与合成头寸

Put-call parity and synthetic positions

## 只记一个等式，再移项

**Remember one equation, then rearrange**

> $c_0 + PV(K) = p_0 + S_0$
>
> Call + Bond = Put + Stock

条件：欧式 call 与 put 使用同一标的、同一执行价和同一到期日；本页公式假设持有标的没有中途收入。左边称 fiduciary call，右边称 protective put。

Conditions: the European call and put have the same underlying, strike and maturity. This form assumes no interim underlying income. The left side is a fiduciary call; the right side is a protective put.

| 到期状态 / State | Call + 到期支付 K 的债券<br>Call + bond paying K | 标的 + Put<br>Stock + put |
|---|---|---|
| $S_T \geqslant K$ | $(S_T-K) + K = S_T$ | $S_T + 0 = S_T$ |
| $S_T < K$ | $0 + K = K$ | $S_T + (K-S_T) = K$ |

两边无论股价如何都支付 max(ST,K)，因此今天价值相等。式子右边的 S0 是现货价格；债券今天只需投入 PV(K)。

Both sides pay max(ST,K) in every terminal state, so they must have the same value today. S0 is today's spot price; the bond costs only PV(K) today.

> Synthetic put: $p_0 = c_0 + PV(K) - S_0$
>
> Synthetic stock: $S_0 = c_0 - p_0 + PV(K)$
>
> Synthetic call: $c_0 = p_0 + S_0 - PV(K)$

正号代表买入，负号代表卖出。Q7 的 put 复制为买 call、买无风险债、卖标的；Q49 的标的复制为买 call、卖 put、买无风险债。

A plus sign means long; a minus sign means short. Q7 replicates a put with a long call, long bond and short underlying. Q49 replicates stock with a long call, short put and long bond.

易错点 / Take care

Q5：p=10+60/1.03-40=28.25。不要把 K=60 直接当作债券今天的价格，也不要把整项再折现。

Q5: p=10+60/1.03-40=28.25. Do not treat the terminal strike payment as today's bond price or discount the entire equation again.

# 看涨看跌平价与合成头寸

Put-call parity and synthetic positions

## 用“远期 + 债券”代替现货

### Replace stock with a forward plus a bond

一份交割价为 F0(T) 的多头远期到期支付 ST-F0(T)。再配一张到期支付 F0(T) 的无风险债券，就得到 ST，等价于持有标的。

A long forward with delivery price F0(T) pays ST-F0(T). Add a risk-free bond paying F0(T) at maturity to obtain ST, matching the underlying payoff.

> $S_0$ = PV[$F_0$(T)] (no income; fair zero-value forward)
>
> $c_0$ + PV(K) = $p_0$ + PV[$F_0$(T)]
>
> $c_0$ - $p_0$ = [$F_0$(T) - K] / (1 + r)^T

## Q8：数字进入正确的位置

### Q8: put each number in the right place

> $p_0$=4; $F_0$(T)=50; K=47; r=10%; T=9/12
>
> $c_0$ = 4 + (50-47)/1.10^0.75 = 6.79

F0(T)=50 是未来交割价格，不是一份公允远期今天值 50。Q27 的合成保护性看跌组合是：买 put、买 forward、买无风险债券；右边债券到期支付应匹配远期交割价 F0(T)。

F0(T)=50 is a delivery price, not a current forward contract value of 50. Q27’s synthetic protective put holds a put, a long forward and a risk-free bond; that bond’s terminal amount must match the forward delivery price F0(T).

| 组合 / Portfolio | 到期支付 / Terminal payoff |
|---|---|
| Fiduciary call | max(ST-K,0) + K |
| Synthetic protective put | max(K-ST,0) + [ST-F0(T)] + F0(T) |
| 共同结果 / Common result | max(ST, K) |

易错点 / Take care

Q47 有措辞问题：题干称 put-call-forward parity，但正确选项 A 写的是普通现货平价。保留原答案 A；学习时必须区分 S0 与 PV[F0(T)]。

Q47 has a wording issue: it names put-call-forward parity, but answer A states spot put-call parity. Retain A for the supplied question while distinguishing S0 from PV[F0(T)].

# 原题 Q05

代表题 / CORE

看涨看跌平价与合成头寸 | Put-call parity and synthetic positions

分析师取得以下资料。根据看涨看跌平价关系，看跌期权的价格最接近多少?

An analyst gathers the following information. According to put-call parity, the price of the put is closest to:

| 项目 Item | 数值 Value |
|---|---|
| 看涨期权价格 Call price | $10 |
| 股票价格 Stock price | $40 |
| 执行价 Exercise price | $60 |
| 利率 Interest rate | 3% |
| 到期时间 Time to expiry | 1 年 / 1 year |

**A** $28.25.

**B** $30.00.

**C** $108.25.

## 参考答案 / Supplied answer A

使用无股息欧式期权平价：看涨期权加执行价的现值，等于看跌期权加股票。把公式移项求 put，先将一年后支付的 $60 折现为 $58.25，再加 call 的 $10、减股票的 $40，得到 $28.25。

Use European put-call parity for a non-dividend-paying stock: the call plus the present value of the strike equals the put plus the stock. Discount the $60 strike due in one year to $58.25, then add the $10 call and subtract the $40 stock price. The put is worth about $28.25.

$$c_0 + K/(1+r)^T = p_0 + S_0$$

$$p_0 = c_0 + K/(1+r)^T - S_0$$

$$p_0 = 10 + 60/1.03 - 40 = \$28.2524$$

易错点 / Take care

B 没有折现执行价；C 错把股票价格加上去了。

B fails to discount the strike; C adds the stock price instead of subtracting it.

# 原题 Q07

看涨看跌平价与合成头寸 | Put-call parity and synthetic positions

根据看涨看跌平价关系，欧式看跌期权的到期收益等价于下列哪个组合的到期收益?

According to put-call parity, the payoff of a European put option is equivalent to a payoff of a portfolio consisting of:

**A** 标的资产多头、看涨期权空头、无风险债券多头。
a long asset, a short call and a long risk-free bond.

**B** 标的资产空头、看涨期权多头、无风险债券多头。
a short asset, a long call and a long risk-free bond.

**C** 标的资产空头、看涨期权空头、无风险债券空头。
a short asset, a short call and a short risk-free bond.

## 参考答案 / Supplied answer B

将平价公式移项可得：put = call - stock + bond。正号代表买入，负号代表卖空，因此需要做空标的、买入 call、买入到期支付执行价的无风险债券。这个组合在两个到期情景下都与 put 相同。

Rearrange put-call parity to obtain put = call - stock + bond. Positive signs indicate long positions and negative signs indicate short positions, giving short stock, long call, and a long risk-free bond paying the strike at maturity. The portfolio matches the put in both terminal price states.

$p_0 = c_0 - S_0 + PV(K)$

If $S_T >= K$: $(S_T - K) - S_T + K = 0$

If $S_T < K$: $0 - S_T + K = K - S_T$

易错点 / Take care

先写平价再移项，避免只凭“看涨/看跌方向”猜组合。债券到期支付金额应为 K。

Write parity and rearrange it before assigning long or short positions. The bond must pay K at maturity.

# 原题 Q08

代表题 / CORE

看涨看跌平价与合成头寸 | Put-call parity and synthetic positions

投资者持有一只无风险债券及一份无股息股票远期合约的多头头寸。远期交割价格为 £50，年无风险利率为 10%。同一股票一份九个月到期、执行价为 £47 的看跌期权价格为 £4。相同期限与执行价的看涨期权价格最接近多少?

An investor takes a long position in a risk-free bond and in a forward contract on a non-dividend-paying stock. The forward contract is priced at £50. The annual risk-free rate is 10%. A nine-month put option on the stock with an exercise price of £47 trades at £4. The price of a nine-month call option on the stock with an exercise price of £47 is closest to:

**A** £6.79.

**B** £7.22.

**C** £7.30.

## 参考答案 / Supplied answer A

使用看涨看跌远期平价：c - p = PV(F_0 - K)。九个月为 0.75 年，远期交割价与执行价之差为 £3，需折现至今天再加上 put 的 £4。计算得到 call 约 £6.79。

Use put-call-forward parity: c - p = PV(F_0 - K). Nine months is 0.75 years, and the difference between the forward delivery price and the strike is £3. Discount that difference to today and add the £4 put price to obtain a call price of about £6.79.

> T = 9/12 = 0.75
>
> $c_0 - p_0 = (F_0 - K)/(1+r)^T$
>
> $c_0 = 4 + (50 - 47)/(1.10^0.75) = GBP\ 6.7930$

易错点 / Take care

£50 是远期交割价格 F_0，不是远期合约初始价值；折现指数是 -0.75。

£50 is the forward delivery price F_0, not the forward contract value at initiation; the discount exponent is -0.75.

# 原题 Q27

看涨看跌平价与合成头寸 | Put-call parity and synthetic positions

根据看涨、看跌与远期的平价关系，信托看涨组合的到期收益等同于下列哪一个组合的到期收益?

According to put-call-forward parity, the payoff on a fiduciary call is equivalent to the payoff on a portfolio consisting of:

**A** 看涨期权多头加无风险债券空头。
a long call and a short risk-free bond.

**B** 看跌期权多头、远期合约多头加无风险债券多头。
a long put, a long forward contract and a long risk-free bond.

**C** 看涨期权空头、远期合约多头加无风险债券多头。
a short call, a long forward contract and a long risk-free bond.

## 参考答案 / Supplied answer B

信托看涨组合为看涨期权加到期支付 X 的无风险债券，到期收益是 max(S_T,X)。另一边用远期多头加到期支付远期交割价 F0 的债券来合成标的资产，再加看跌期权，也得到 max(S_T,X)。因此选择 B。

A fiduciary call combines a call with a risk-free bond paying X at expiry, producing max(S_T,X). On the other side, a long forward plus a bond paying the forward delivery price F0 replicates the underlying; adding a put again gives max(S_T,X). Therefore B is correct.

Call + bond paying X: $\max(S_T\text{-}X,0) + X = \max(S_T,X)$.

Put + long forward + bond paying $F_0$: $\max(X\text{-}S_T,0) + (S_T\text{-}F_0) + F_0 = \max(S_T,X)$.

易错点 / Take care

两个组合中的债券到期面额未必相同：左边是 X，右边是 F0。

The bond faces need not match: the left-hand bond pays X, while the right-hand bond pays F0.

审校说明 / Wording and scope

题干省略了债券面额；这里补足为与各组合现金流相匹配的到期支付金额。

The question omits bond face values; the explanation specifies the maturity payments required to match each portfolio.

# 原题 Q47

看涨看跌平价与合成头寸 | Put-call parity and synthetic positions

以下哪一项最准确地描述看涨-看跌-远期平价关系（put-call-forward parity）？

Which of the following best describes put-call-forward parity?

**A** 行权价的现值加看涨期权价格，等于看跌期权价格加标的资产价格。
The present value of the exercise price plus the call price equals the put price plus the underlying price.

**B** 标的资产价格加看涨期权价格，等于行权价的现值加看跌期权价格。
The underlying price plus the call price equals the present value of the exercise price plus the put price.

**C** 看涨期权价格减看跌期权价格，等于行权价的现值减标的资产价格。
The call price minus the put price equals the present value of the exercise price minus the underlying price.

## 参考答案 / Supplied answer A

按照题目选项及所给答案，正确关系是“看涨期权 + 行权价现值 = 看跌期权 + 现货”。这是同到期日、同行权价的欧式期权在标的不支付收益等标准条件下的现货看涨-看跌平价。若改用远期价格表达，应写为看涨减看跌等于远期价格与行权价差额的现值。

Following the answer choices and supplied key, the correct relationship is call plus the present value of the strike equals put plus spot. This is ordinary spot put-call parity for European options with the same strike and expiry, under the standard no-income assumptions. In forward-price form, call minus put equals the present value of the difference between the forward price and strike.

$$c_0 + PV(K) = p_0 + S_0$$

$$c_0 - p_0 = PV(F_0 - K)$$

易错点 / Take care

不要把公式移项后的正负号记反：c - p = S - PV(K)，不是PV(K) - S。

Keep the signs correct when rearranging: c - p = S - PV(K), not PV(K) - S.

审校说明 / Wording and scope

原题题干及答案解析把现货平价称为put-call-forward parity，措辞不够精确。保留原题与答案A，同时区分现货形式和远期形式；不把F_0直接当作S_0。

The original stem and solution label the spot relationship as put-call-forward parity. The wording is imprecise. Preserve answer A, but distinguish the spot form from the forward form; F_0 is not S_0.

# 原题 Q49

代表题 / CORE

看涨看跌平价与合成头寸 | Put-call parity and synthetic positions

根据看涨-看跌平价，以下哪一种组合等价于持有标的资产的多头头寸?

Based on put-call parity, which of the following is equivalent to a long position in the underlying asset?

**A** 买入看涨期权、买入看跌期权、卖空债券。
Long call, long put, and short bond

**B** 买入看跌期权、卖出看涨期权、买入债券。
Long put, short call, and long bond

**C** 买入看涨期权、卖出看跌期权、买入债券。
Long call, short put, and long bond

## 参考答案 / Supplied answer C

由c + PV(K) = p + S移项可得S = c - p + PV(K)。因此，买入看涨期权、卖出看跌期权，再买入一张到期支付K的无风险零息债券，可以复制标的资产。两种期权须为同到期日、同行权价的欧式期权，且采用标准无收益情形。

Rearranging c + PV(K) = p + S gives S = c - p + PV(K). A long call, short put, and long risk-free zero-coupon bond paying K at expiry therefore replicate the underlying. The European options must share the same strike and expiry, under the standard no-income setup.

$S_0 = c_0 - p_0 + PV(K)$

At expiry: $\max(S_T - K,0) - \max(K - S_T,0) + K = S_T$

易错点 / Take care

债券的到期面值必须是K；“买看涨、卖看跌”本身复制的是到期支付S_T - K，而非完整的S_T。

The bond must pay K at expiry. A long call and short put alone produce S_T - K, rather than the full S_T.

# 二叉树、风险中性概率与复制

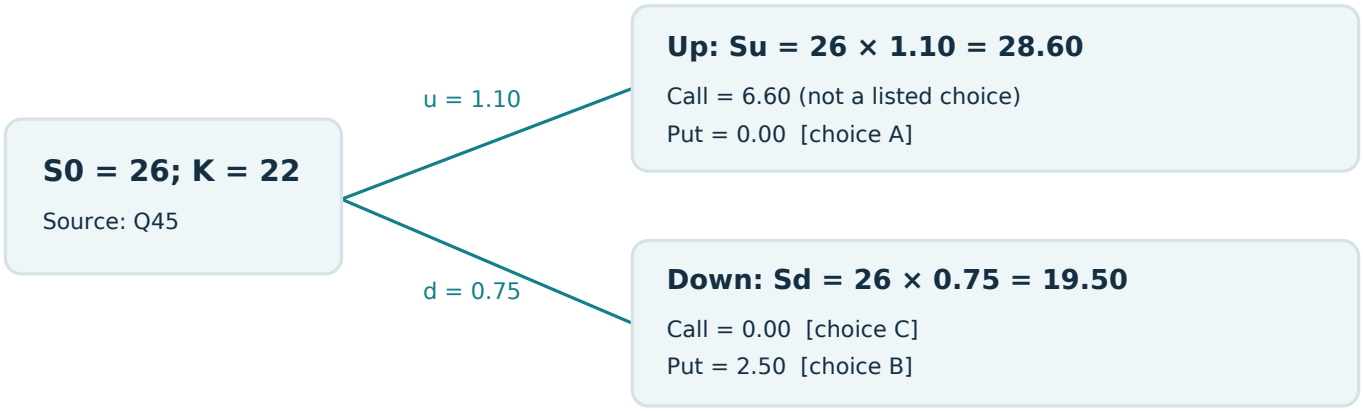
Binomial trees, risk-neutral probabilities and replication

## 先做终点 payoff，再谈今天的价格

### Compute terminal payoffs before today's price

一期模型只允许一个上行状态和一个下行状态。u、d 是总回报倍数：上涨 10% 是 u=1.10；下跌 25% 是 d=0.75。

A one-period model has an up state and a down state. u and d are gross return factors: an increase of 10% gives u=1.10; a decrease of 25% gives d=0.75.




Q45 只比较三个列出选项的终点 payoff，不需要风险中性概率或无风险利率；下跌后的 put payoff 为 2.50，选 B。

Q45 compares the terminal payoffs of the three listed choices, so no risk-neutral probability or risk-free rate is needed. The down-state put pays 2.50, giving B.

## 风险中性概率：让预期增长率等于无风险利率

### Risk-neutral weights match risk-free expected growth

> R = (1 + r)^T; p_up = (R-d)/(u-d)
>
> p_up = [$S_0$ × R - Sd]/(Su-Sd); p_down = 1-p_up

使用条件是 d<R<u，使概率处于 0 与 1 之间。这是无套利估值权重，不是对真实上涨概率的预测；不需要投资者风险厌恶程度或标的实际预期收益率。

The no-arbitrage condition d<R<u places the probability between 0 and 1. These are valuation weights, not forecasts of actual up probabilities. Investor risk aversion and the asset's real-world expected return are not required.

易错点 / Take care

Q50：p_up=(16×1.04-12)/(22-12)=0.464。Q51：p_up=(1.05-0.75)/(1.25-0.75)=0.60，但题目问下降，所以答 0.40。

Q50 gives p_up=0.464. Q51 gives p_up=0.60, but asks for a decrease, so the answer is 0.40.

# 二叉树、风险中性概率与复制

Binomial trees, risk-neutral probabilities and replication

## 方法一：期望 payoff 再折现

**Method 1: discount the weighted terminal payoff**

$$V_0 = [p_up \times Vu + (1-p_up) \times Vd] / R$$

Vu、Vd 必须是期权在各状态的 payoff，不是标的价格。折现只做一次。求真实价格仍需要足够的利率和时间信息；不能给 Q45 擅自补一个利率。

Vu and Vd must be option payoffs, not underlying prices. Discount once. Valuation requires enough rate and timing information; no rate should be invented for Q45.

## 方法二：股票份数 + 无风险账户

**Method 2: underlying units plus a risk-free account**

$$Delta = (Vu - Vd)/(Su - Sd)$$

$$B0 = (Vd - Delta \times Sd)/R$$

$$V_0 = Delta \times S_0 + B0$$

Delta 表示每份期权需要多少单位标的。普通 call 的复制通常是买入部分股票并借款；普通 put 的复制通常是卖空部分股票并持有无风险资产。

Delta gives the underlying units required per option. A vanilla call is usually replicated with a fraction of the stock plus borrowing; a vanilla put with a short stock position plus a risk-free investment.

| 考点 / Test | 判断 / Rule | 原题 / Q |
|---|---|---|
| 参数 u 上升，r 与 d 不变<br>u rises with r and d fixed | 分母增大，p_up 下降<br>Larger denominator; lower p_up | 54 |
| 定价是否需要真实预期收益率?<br>Need actual expected return? | 不需要；需要无风险利率<br>No; the risk-free rate is required | 52 |
| 跨多个时点复制期权<br>Replication across dates | 随状态改变重调 hedge ratio<br>Rebalance the hedge ratio as states change | 44 |

易错点 / Take care

Q44 原答 A 的“根据行权可能性调整”是直观表述。严格说应更新复制比例 Delta；Delta 与风险中性上涨概率、实际行权概率不是同一个量。

Q44’s answer A describes adjustment in terms of exercise likelihood. More precisely, update the replicating hedge ratio Delta. Delta, a risk-neutral up probability and an actual exercise probability are different quantities.

# 原题 Q44

二叉树、风险中性概率与复制 | Binomial trees, risk-neutral probabilities and replication

关于看涨期权的复制策略，以下哪一种说法最准确?

Which of the following is most accurate regarding a call option replication strategy?

**A** 策略需要在期权存续期内，随着行权可能性的变化进行调整。

The strategy requires adjustment over the life of the option contract based on the likelihood of exercise

**B** 若看涨期权被行权，策略需要用贷款的资金买入标的资产。

If the call option is exercised, the strategy requires purchasing the underlying from the proceeds of the loan

**C** 合约起始时，策略需要买入标的的远期合约并按无风险利率借款。

At inception, the strategy requires buying long a forward contract on the underlying and borrowing at the risk free rate

## 参考答案 / Supplied answer A

看涨期权收益非线性，复制它通常需要持有一定数量的标的资产，并以无风险借贷账户融资。随着标的价格和剩余时间变化，复制所需的持股数量 delta 也变化，因此要动态调整，A 最准确。静态买入一份远期只能产生线性收益，不能直接复制看涨期权。

A call has a nonlinear payoff and is typically replicated with a quantity of the underlying financed through a risk-free borrowing or lending account. The required stock holding, delta, changes with the underlying price and remaining time, so the strategy must be rebalanced, making A best. A static long forward only creates a linear payoff and cannot by itself replicate a call.

> One-period call hedge ratio: delta = $(c_u-c_d)/(S_u-S_d)$.
>
> Replicating call: c = delta*S + risk-free account.

易错点 / Take care

不要把 delta 直接当作真实世界的行权概率，也不要认为复制永远只需买入一股股票。

Do not equate delta with the real-world exercise probability or assume replication always requires exactly one share.

审校说明 / Wording and scope

原选项使用“根据行权可能性调整”的直观表述；更精确的执行规则是依据模型的对冲比率 delta 调仓。

The source option uses exercise likelihood as intuition; the more precise trading rule is to rebalance according to the model hedge ratio, delta.

# 原题 Q45

二叉树、风险中性概率与复制 | Binomial trees, risk-neutral probabilities and replication

某分析师收集了下表信息。根据单期二叉树定价模型，以下哪一项的到期支付最大?

An analyst collects the following information. Based on a one-period binomial pricing model, which of the following has the largest payoff?

| 项目 / Item | 数值 / Value |
|---|---|
| 当前股价 / Current stock price | €26 |
| 上涨情景总回报倍数 / Gross return from an up move | 1.10 |
| 下跌情景总回报倍数 / Gross return from a down move | 0.75 |
| 看涨与看跌期权的行权价 / Call and put exercise price | €22 |

**A** 上涨情景下的看跌期权
Put option following an up move

**B** 下跌情景下的看跌期权
Put option following a down move

**C** 下跌情景下的看涨期权
Call option following a down move

## 参考答案 / Supplied answer B

先计算两种情景的股价，再代入各自期权的支付函数。上涨时股价为€28.60，看跌期权支付为0；下跌时股价为€19.50，看跌期权支付为€2.50，而看涨期权支付为0。因此，B在三个选项中的支付最大。

First calculate the stock price in each state, then apply the relevant option payoff. In the up state, the stock is €28.60 and the put pays zero. In the down state, the stock is €19.50, so the put pays €2.50 and the call pays zero. B therefore has the largest payoff among the three choices.

$S_u = 26 \times 1.10 = 28.60$; $S_d = 26 \times 0.75 = 19.50$

$p_u = \max(22 - 28.60, 0) = 0$

$p_d = \max(22 - 19.50, 0) = 2.50$

$c_d = \max(19.50 - 22, 0) = 0$

易错点 / Take care

只比较题目列出的三个选项；未列出的“上涨情景看涨期权”支付虽更高，却不是可选答案。这里求支付，不需要风险中性概率。

Compare only the three listed choices. The up-state call has a larger payoff, but it is not an answer choice. Payoffs do not require risk-neutral probabilities.

# 原题 Q50

代表题 / CORE

二叉树、风险中性概率与复制 | Binomial trees, risk-neutral probabilities and replication

某分析师收集了下列标的资产信息。使用单期二叉树模型，价格上涨的风险中性概率最接近多少？

An analyst gathers the following information about an underlying. Using a one-period binomial model, the risk-neutral probability of a price increase is closest to:

| 项目 / Item | 数值 / Value |
|---|---|
| 当前标的资产价格 / Current underlying price | $16.0 |
| 期末上涨情景价格 / End-of-period up price | $22.0 |
| 期末下跌情景价格 / End-of-period down price | $12.0 |
| 无风险利率 / Risk-free rate | 4.0% |

**A** 0.38.

**B** 0.46.

**C** 0.54.

## 参考答案 / Supplied answer B

风险中性概率使标的资产的风险中性预期总回报等于无风险总回报。在本题单期利率为4%的口径下，令16×1.04 = q×22 + (1-q)×12，解得上涨概率q = 0.464，最接近0.46。

The risk-neutral probability makes the underlying's expected gross return, under risk-neutral weighting, equal to the risk-free gross return. Treating 4% as the rate for this period, solve 16 × 1.04 = q × 22 + (1 - q) × 12. The up probability is 0.464, closest to 0.46.

$$q = [S_0 \times (1+r) - S_d] / (S_u - S_d)$$

$$q = (16 \times 1.04 - 12) / (22 - 12) = 0.464$$

易错点 / Take care

风险中性概率是无套利定价权重，不是分析师预测的真实上涨概率；本题问上涨，不是1-q。

A risk-neutral probability is a no-arbitrage pricing weight, not a forecast of the actual chance of an increase. The question asks for q, not 1 - q.

审校说明 / Wording and scope

原答案说明中有一处把“标的资产期末价格的折现加权平均”写成等于当前期权价格；准确说应等于当前标的资产价格。期权价格则由期权支付的折现加权平均得到；原选择答案不受影响。

The original explanation states that the discounted weighted underlying prices equal the current option price. They should equal the current underlying price. The option price is the discounted weighted option payoff; the keyed answer is unaffected.

# 原题 Q51

代表题 / CORE

二叉树、风险中性概率与复制 | Binomial trees, risk-neutral probabilities and replication

投资者预计某标的资产在未来一年会上涨或下跌25%。若无风险利率为5%，标的资产价格下跌的风险中性概率最接近多少?

An investor expects an underlying to go up or down by 25% over the next year. If the risk-free rate is 5%, the risk-neutral probability of a decrease in the underlying price is closest to:

**A** 0.25.

**B** 0.40.

**C** 0.60.

## 参考答案 / Supplied answer B

上涨和下跌的总回报倍数分别为1.25和0.75。先计算上涨的风险中性概率q = (1.05 - 0.75)/(1.25 - 0.75) = 0.60，再用1减去q。题目所问的下跌概率为0.40。

The up and down gross-return multipliers are 1.25 and 0.75. First calculate the risk-neutral up probability: q = (1.05 - 0.75)/(1.25 - 0.75) = 0.60. The question asks for the down probability, which is 1 - q = 0.40.

> u = 1.25; d = 0.75; R = 1.05
>
> q_up = (R - d) / (u - d) = 0.60
>
> q_down = 1 - q_up = 0.40

易错点 / Take care

看到“上涨/下跌25%”不能直接认为概率各50%；上涨概率算出0.60后还要转换成题目要求的下跌概率。

A symmetric ±25% move does not imply 50/50 risk-neutral probabilities. After obtaining 0.60 for an increase, convert it to the requested decrease probability.

# 原题 Q52

二叉树、风险中性概率与复制 | Binomial trees, risk-neutral probabilities and replication

使用二叉树模型时，以下哪一项因素会影响期权价格?

Which of the following factors affects the option price when using a binomial model? The:

**A** 无风险利率。
risk-free rate.

**B** 投资者的风险厌恶程度。
level of investors' risk aversion.

**C** 标的资产的预期回报。
expected return of the underlying.

## 参考答案 / Supplied answer A

二叉树无套利定价使用无风险利率计算风险中性概率，并把风险中性期望支付折现到今天。在给定当前价格和上涨/下跌情景的情况下，不需要输入投资者的风险厌恶程度或标的资产的真实预期回报。

Binomial no-arbitrage pricing uses the risk-free rate to determine risk-neutral probabilities and discount the risk-neutral expected payoff. Given today's price and the up/down states, it does not require investors' risk aversion or the underlying's actual expected return as inputs.

> Option value = [q × payoff_u + (1 - q) × payoff_d] / R

易错点 / Take care

"不需要真实预期回报"不等于"期权没有风险"，而是无套利复制已把风险定价问题转换成定价权重。

Not needing the actual expected return does not mean the option is risk-free. No-arbitrage replication converts the valuation problem into pricing weights.

# 原题 Q54

二叉树、风险中性概率与复制 | Binomial trees, risk-neutral probabilities and replication

其他条件不变，在单期二叉树模型中，若上涨情景的总回报倍数提高，资产价格上涨的风险中性概率将如何变化?

All else being equal, if the up gross return increases in a one-period binomial model, the risk-neutral probability of an upward price movement of the asset will:

**A** 下降。
decrease.

**B** 保持不变。
remain the same.

**C** 上升。
increase.

## 参考答案 / Supplied answer A

风险中性上涨概率为q = (R - d)/(u - d)。在无风险总回报R与下跌倍数d不变、且满足无套利条件d < R < u时，上涨倍数u增大使分母增大，因此q下降。更大的上涨幅度需要更小的上涨权重，才能保持风险中性期望回报为R。

The risk-neutral up probability is q = (R - d)/(u - d). Holding the risk-free gross return R and down multiplier d fixed, and assuming d < R < u, a larger u increases the denominator and reduces q. A larger upside move requires a smaller up-state weight to keep the risk-neutral expected gross return equal to R.

> q = (R - d) / (u - d)
>
> R and d fixed: u increases -> q decreases

### 易错点 / Take care

本题问定价权重，不能凭“上涨空间更大所以更可能上涨”的直觉作答。

This is a question about pricing weights. A larger possible gain does not imply a larger risk-neutral probability.

### 审校说明 / Wording and scope

原答案说明中有一处把“标的资产期末价格的折现加权平均”写成等于当前期权价格；准确说应等于当前标的资产价格。期权价格则由期权支付的折现加权平均得到；原选择答案不受影响。

The original explanation states that the discounted weighted underlying prices equal the current option price. They should equal the current underlying price. The option price is the discounted weighted option payoff; the keyed answer is unaffected.

# 公式速查 1：远期与利率

Formula sheet 1: forwards and rates

## 先统一口径

### Use consistent units and conventions

S 表示标的价格；K 表示已锁定交割价或期权执行价；N 在远期中为单位数，在 FRA/互换中为名义本金。T 为年数，α 为计息年数，r 与 L 必须用小数。

S denotes the asset price; K is a locked delivery price or an option strike. N is underlying units for a forward and notional principal for an FRA or swap. T and α are year fractions; r and L are decimal rates.

| 用途 / Use | 公式 / Formula | 原题 / Q |
|---|---|---|
| 无收入远期价格<br>Forward price | F0 = S0(1+r)^T | 30, 38 |
| 远期多头现值<br>Long forward value | Vt = N[St - K/(1+r)^(T-t)] | 65 |
| 远期空头到期价值<br>Short terminal value | VT = N(K-ST) | 55 |
| 股息连续收益率<br>Continuous dividend yield | F0 = S0 exp[(r-q)T] | 37, 60 |
| 到期支付成本 CT<br>Cost paid at maturity | F0 = S0(1+r)^T + CT | 42 |
| 连续持有成本<br>Continuous carry | F0 = S0 exp[(r+u-q-y)T] | 21, 38 |
| 零息债隐含远期利率<br>Implied forward rate | f = [P(0,t)/P(0,T)]^(1/(T-t))-1 | 6 |
| FRA 多头期初结算<br>Long FRA advance settlement | N(L-K)α / (1+Lα) | 31, 62 |
| 互换固定支付方<br>Swap fixed payer | N(L-K)α | 46 |
| 互换固定收取方<br>Swap fixed receiver | N(K-L)α | 41 |

易错点 / Take care

每道题都检查：年复利还是连续复利？日数除以 365 还是其他基准？费用期初付还是期末付？答案要每股、每点还是总金额？

For every calculation, check annual versus continuous compounding, the day-count basis, cost timing, and whether the answer is per unit or for the entire position.

# 公式速查 2：期权与二叉树

Formula sheet 2: options and binomial models

## 到期 payoff 与利润

**Terminal payoff and profit**

| 头寸 / Position | Payoff | Profit |
|---|---|---|
| Long call | max(ST-K,0) | Payoff-c0 |
| Short call | -max(ST-K,0) | Payoff+c0 |
| Long put | max(K-ST,0) | Payoff-p0 |
| Short put | -max(K-ST,0) | Payoff+p0 |

## 平价与边界：欧式、同 K 同 T、无中途收入

**Parity and bounds: European, same K and T, no interim income**

$c_0 + PV(K) = p_0 + S_0$

$c_0 - p_0 = PV[F_0(T) - K]$

$\max[0, S_0\text{-}PV(K)] \le c_0 \le S_0$

$\max[0, PV(K)\text{-}S_0] \le p_0 \le PV(K)$

## 一期二叉树

**One-period binomial tree**

Su=S0u; Sd=S0d; R=(1+r)^T

p_up=(R-d)/(u-d)=[S0R-Sd]/(Su-Sd)

p_down=1-p_up; $V_0$=[p_up Vu+(1-p_up)Vd]/R

Delta=(Vu-Vd)/(Su-Sd); B0=(Vd-Delta Sd)/R; $V_0$=Delta $S_0$+B0

易错点 / Take care

口诀：payoff 不含 premium；price 不等于 value；put 可涨但时间效应不绝对；probability 不等于 Delta。

Recall: payoff excludes premium; delivery price differs from contract value; put maturity effects are conditional; probability is not Delta.

# 参考答案与定位：Q1-Q33

Answer key and locations: Q1-Q33

此表保留原参考答案字母。Q47、Q50、Q54、Q59、Q62 等题的文字或口径说明见对应题页及审校附录。

The supplied answer letters are retained. See the individual pages and editorial appendix for wording or scope notes on Q47, Q50, Q54, Q59, Q62 and others.

| 原题 / Q | 答案 | 单元 | 本书页 | 题目PDF页 | 答案PDF页 |
|---|---|---|---|---|---|
| Q01 | C | U7 | 61 | 134 | 1 |
| Q02 | C | U7 | 62 | 134 | 2 |
| Q03 | B | U7 | 63 | 134 | 3 |
| Q04 | A | U6 | 50 | 134 | 4 |
| Q05 | A | U8 | 69 | 135 | 5 |
| Q06 | B | U5 | 42 | 135 | 6 |
| Q07 | B | U8 | 70 | 135 | 7 |
| Q08 | A | U8 | 71 | 135 | 8 |
| Q09 | C | U1 | 5 | 135 | 9 |
| Q10 | A | U2 | 16 | 136 | 10 |
| Q11 | A | U7 | 64 | 136 | 11 |
| Q12 | B | U1 | 6 | 136 | 12 |
| Q13 | B | U1 | 7 | 136 | 13 |
| Q14 | B | U2 | 17 | 136 | 14 |
| Q15 | C | U1 | 8 | 136 | 15 |
| Q16 | B | U6 | 51 | 137 | 16 |
| Q17 | A | U1 | 9 | 137 | 17 |
| Q18 | A | U2 | 18 | 137 | 18 |
| Q19 | C | U2 | 19 | 137 | 19 |
| Q20 | C | U2 | 20 | 137 | 20 |
| Q21 | A | U3 | 24 | 137 | 21 |
| Q22 | B | U4 | 35 | 138 | 22 |
| Q23 | C | U6 | 52 | 138 | 23 |
| Q24 | A | U3 | 25 | 138 | 24 |
| Q25 | B | U2 | 21 | 138 | 25 |
| Q26 | A | U6 | 53 | 138 | 26 |
| Q27 | B | U8 | 72 | 138 | 27 |
| Q28 | C | U1 | 10 | 139 | 28 |
| Q29 | B | U5 | 43 | 139 | 29 |
| Q30 | A | U3 | 26 | 139 | 30 |
| Q31 | C | U5 | 44 | 139 | 31 |
| Q32 | A | U1 | 11 | 139 | 32 |
| Q33 | A | U5 | 45 | 139 | 33 |

# 参考答案与定位：Q34-Q65

Answer key and locations: Q34-Q65

此表保留原参考答案字母。Q47、Q50、Q54、Q59、Q62 等题的文字或口径说明见对应题页及审校附录。

The supplied answer letters are retained. See the individual pages and editorial appendix for wording or scope notes on Q47, Q50, Q54, Q59, Q62 and others.

| 原题 / Q | 答案 | 单元 | 本书页 | 题目PDF页 | 答案PDF页 |
|---|---|---|---|---|---|
| Q34 | B | U7 | 65 | 140 | 34 |
| Q35 | A | U1 | 12 | 140 | 35 |
| Q36 | C | U4 | 36 | 140 | 36 |
| Q37 | B | U3 | 27 | 140 | 37 |
| Q38 | B | U3 | 28 | 140 | 38 |
| Q39 | C | U4 | 37 | 141 | 39 |
| Q40 | C | U4 | 38 | 141 | 40 |
| Q41 | A | U5 | 46 | 141 | 41 |
| Q42 | A | U3 | 29 | 141 | 42 |
| Q43 | B | U3 | 30 | 141 | 43 |
| Q44 | A | U9 | 77 | 142 | 44 |
| Q45 | B | U9 | 78 | 142 | 45 |
| Q46 | B | U5 | 47 | 142 | 46 |
| Q47 | A | U8 | 73 | 142 | 47 |
| Q48 | B | U6 | 54 | 143 | 48 |
| Q49 | C | U8 | 74 | 143 | 49 |
| Q50 | B | U9 | 79 | 143 | 50 |
| Q51 | B | U9 | 80 | 143 | 51 |
| Q52 | A | U9 | 81 | 143 | 52 |
| Q53 | B | U1 | 13 | 144 | 53 |
| Q54 | A | U9 | 82 | 144 | 54 |
| Q55 | B | U3 | 31 | 144 | 55 |
| Q56 | B | U6 | 55 | 144 | 56 |
| Q57 | C | U6 | 56 | 144 | 57 |
| Q58 | C | U6 | 57 | 145 | 58 |
| Q59 | A | U6 | 58 | 145 | 59 |
| Q60 | A | U3 | 32 | 145 | 60 |
| Q61 | C | U6 | 59 | 145 | 61 |
| Q62 | B | U4 | 39 | 145 | 62 |
| Q63 | A | U7 | 66 | 146 | 63 |
| Q64 | B | U1 | 14 | 146 | 64 |
| Q65 | C | U3 | 33 | 146 | 65 |

# 做完以后：把错题变成下一轮任务

Review: convert errors into your next study task

## 五类错因，比只统计正确率有用

**Classify the mistake, not just the score**

| 错因 / Error type | 出现时怎么修复 / Corrective action |
|---|---|
| 概念 / Concept | 重读本单元第一页，并用一句话说清定义。<br>Reread the unit’s concept page and state the definition. |
| 方向 / Sign or position | 先写 long/short、call/put、收什么减付什么。<br>Write the position and received-minus-paid cash flows. |
| 公式 / Formula | 只凭空白纸重写公式，再解释每个符号。<br>Reproduce the formula and explain every symbol. |
| 计算 / Arithmetic | 检查年数、百分数、复利、折现与乘数。<br>Check years, decimals, compounding, discounting and units. |
| 审题 / Wording | 圈出 most likely、payoff、profit、value、up/down。<br>Mark the qualifier and the quantity actually requested. |

## 错题记录模板

**A compact error log**

| 题号 / Q | 我的错误 / My mistake | 应记规则 / Rule to retain | 复做 / Retry |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

第一轮先完成 22 道代表题；第二轮完成本单元其余原题。答对但靠猜、超时或解释不清，也归入复习。隔一轮重新做错题时先遮住同页答案。

First complete the 22 core examples, then the remaining supplied questions in each unit. Review guessed, slow and poorly explained correct answers too. Cover the same-page solution when retrying.

易错点 / Take care

自测通过标准：你能说出为什么其他选项错，并在不看讲义时写出计算式。这里的单元顺序与代表题是学习安排，不是考试命中率预测。

Mastery means explaining why alternatives fail and reproducing the calculation without the guide. The order and core examples are study choices, not predictions of exam frequency.

# 原题与原解：需要保留的限定

Editorial notes: limits in the supplied wording

原题编号、数值、三个选项和原参考答案均保留。下列项目已在对应题页注明，避免把简化表述学成无条件规律。

The original numbering, data, three choices and supplied answer letters are retained. The following qualifications are also noted on the relevant question pages.

| 题号 / Q | 核对结论 / Editorial finding |
|---|---|
| 1, 2 | 欧式期权期限效应有条件；Q2 原解已承认 put 期限关系可正可负。<br>European maturity effects are conditional; Q2’ s solution already allows either sign for puts. |
| 27 | 合成现货的债券支付 F0(T)；fiduciary call 的债券支付 K。<br>The synthetic-stock bond pays F0(T); the fiduciary-call bond pays K. |
| 37, 38 | Q37 原解采用连续复利；Q38 的净持有成本项不含融资利率 r。<br>Q37 uses continuous compounding; Q38’ s net carry term excludes financing. |
| 44 | 实际复制按 Delta 调整；不能把 Delta 与行权概率等同。<br>Replication adjusts Delta, which is not the exercise probability. |
| 47 | 题干写远期平价，选项 A 实际是现货平价；保留 A 并列出准确公式。<br>The prompt names forward parity but A states spot parity. A is retained with the distinction. |
| 50, 54 | 原解把标的终值的折现期望称为期权价格；准确应为标的现价。<br>Discounted expected underlying value gives spot, not option value. |
| 59 | 按字面并非严格唯一；A 的意图是两条腿都看涨，组合2的净值也可能上涨。<br>Literal wording is not strictly unique. A has two bullish legs, while portfolio 2 can also rise. |
| 62 | 原答 B；此处比较 FRA 提前结算金额的曲率。<br>Answer B compares the curvature of advance FRA settlement. |

易错点 / Take care

Q59 的反例推导：若远期交割价与 put 执行价同为 K，远期多头 + put 多头的到期 payoff = ST-K+max(K-ST,0)=max(ST-K,0)，也具有看涨方向。

For Q59, if the forward delivery price and put strike both equal K, long forward plus long put pays ST-K+max(K-ST,0)=max(ST-K,0), which is also bullish.

# 核心术语中英速查

Core bilingual vocabulary

| English | 中文 | 读题提示 / Reading cue |
|---|---|---|
| underlying | 标的资产 | 用于确定衍生品价值的资产或利率<br>Asset or rate driving the derivative |
| forward commitment | 远期承诺 / 确定承诺 | 双方有履约义务 / Both parties are bound |
| contingent claim | 或有索取权 | 支付取决于条件 / Contingent payoff |
| exercise / strike price | 执行价 / 行权价 | K；原解也用 X / Also written X |
| premium | 权利金 | 期权的购买价格 / Price paid for an option |
| payoff | 到期支付 / 到期收益 | 通常不含原始权利金 / Excludes initial premium |
| profit | 净利润 | 计入权利金 / Includes premium |
| intrinsic / exercise value | 内在价值 / 行权价值 | 立刻行权的价值 / Immediate exercise value |
| time value | 时间价值 | 价格减内在价值 / Price minus intrinsic value |
| moneyness | 实值 / 平值 / 虚值状态 | 比较 S 与 K / Compare S with K |
| cost of carry | 持有成本 | 检查是否包括融资 / Check inclusion of financing |
| convenience yield | 便利收益 | 拥有实物带来的利益 / Benefits of physical ownership |
| variation margin | 变动保证金 | 每日盈亏现金结算 / Daily cash settlement |
| market reference rate (MRR) | 市场参考利率 | 浮动腿使用的参考利率 / Floating-rate reference |
| notional principal | 名义本金 | 用于算利息，并非一定交换 / Calculation base, not necessarily exchanged |
| fiduciary call | 信托看涨组合 | Call + bond paying K |
| protective put | 保护性看跌组合 | Stock + put |
| risk-neutral probability | 风险中性概率 | 无套利估值权重 / Valuation weight |
| hedge ratio / Delta | 对冲比率 | 每份期权所需标的单位数 / Underlying units per option |

# 来源、范围与编写说明

Sources, coverage and preparation notes

## 主要来源：你提供的两份文件

### Primary sources: your two supplied PDFs

题目：《1000题(3).pdf》PDF 第 134-146 页，章节内部页码 1-13，原题 Q1-Q65。答案：《衍生品(1).pdf》PDF 第 1-65 页，每页对应同编号题目。每道题页底均列出两份文件的 PDF 页码。

Question PDF: pages 134-146, section pages 1-13, Q1-Q65. Answer PDF: pages 1-65, one corresponding answer per page. Each question footer records both source PDF pages.

来源类别统一标注为“用户提供题库”。截图不能独立证明每题属于 CFA 官方题、历年真题或某模拟题，因此本讲义不擅自给出该标签。9 个单元是按知识点重组的教学结构，并非官方章节编号。

All questions are labeled as a user-supplied question bank. The screenshots do not independently establish official, past-exam or mock provenance. The nine units are an instructional regrouping, not official chapter numbers.

## 补充核对资料：CFA Institute 2026 公开阅读摘要

### Supplementary checks: CFA Institute 2026 public refresher pages

这些资料用于核对概念边界与主题名称。讲解、推导和习题解析以用户文件为主，未新增模拟选择题。以下标题可点击打开。核对日期：2026-11-02。

These pages support terminology and conceptual boundaries. The teaching and worked solutions primarily use the supplied files; no new mock multiple-choice questions are added. Titles below are clickable. Checked on 2026-11-02.

**R1** Pricing and Valuation of Forward Contracts and for an Underlying with Varying Maturities

**R2** Pricing and Valuation of Futures Contracts

**R3** Pricing and Valuation of Options

**R4** Option Replication Using Put-Call Parity

**R5** Valuing a Derivative Using a One-Period Binomial Model

**R6** Arbitrage, Replication, and the Cost of Carry in Pricing Derivatives

易错点 / Take care

这是围绕所给 65 题的完整双语精讲，不等同于整个 CFA 衍生品考纲的穷尽式教材。英文题干已整理换行和表格，明显的 S40 排印已按原答案修正为 $40；没有更改题目数值或选项。

This is a complete bilingual guide to the supplied 65 questions, not an exhaustive replacement for the entire derivatives curriculum. Line breaks and tables are normalized; the obvious S40 typo is rendered as $40 consistently with the solution. Numerical data and answer choices are preserved.